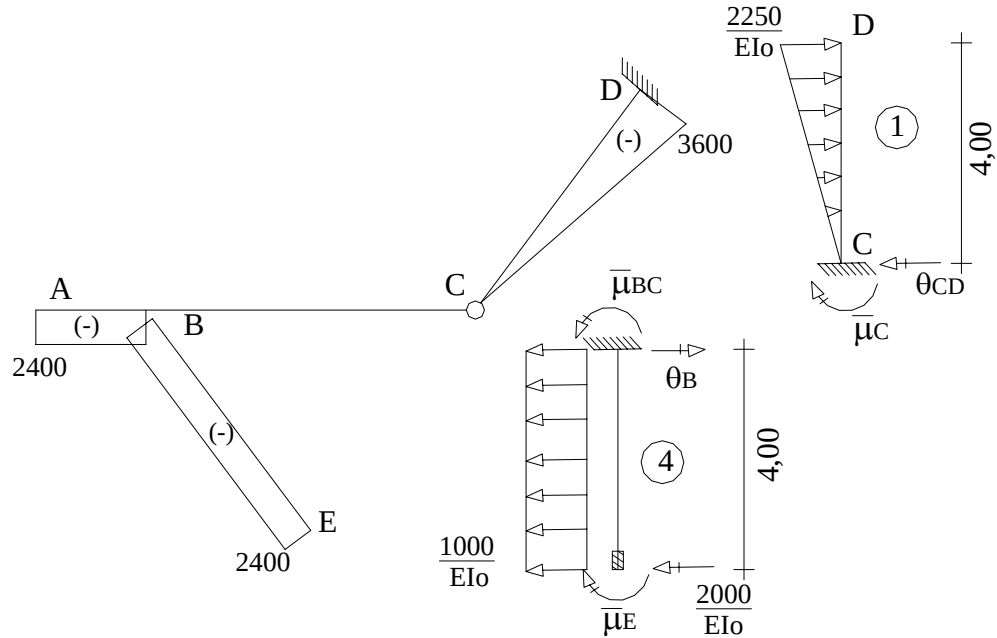
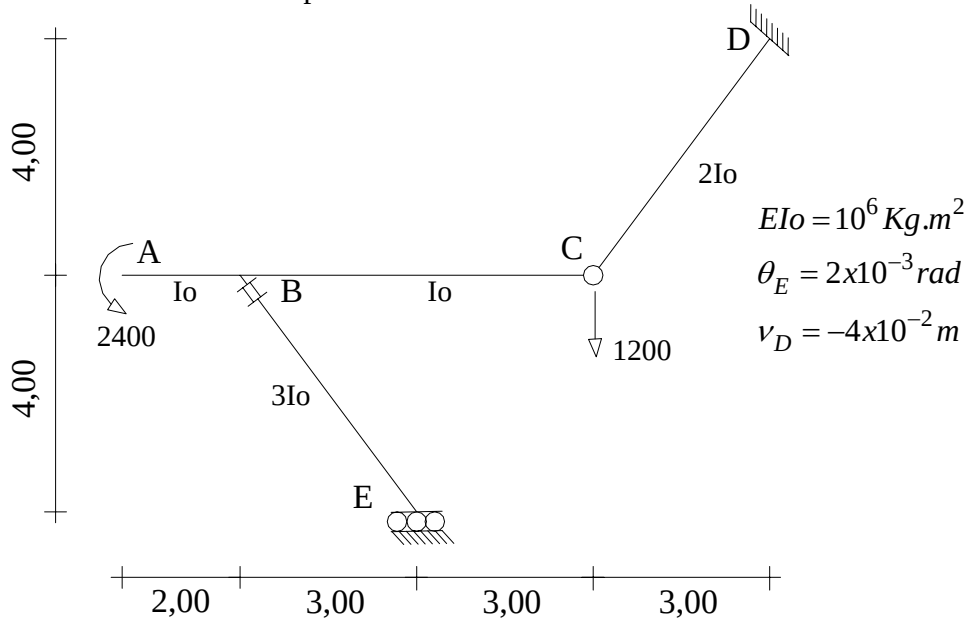


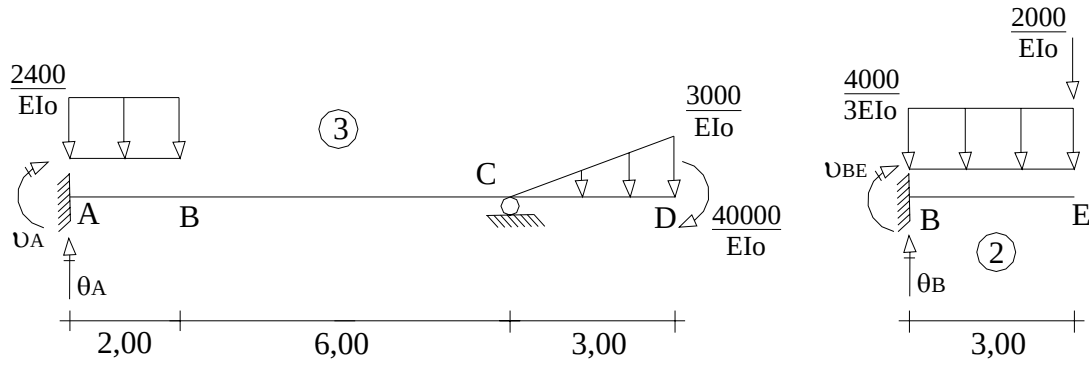
CAPITULO II / III Y IV

TRABAJOS VIRTUALES / VIGA CONJUGADA.

1.- Determinar Elásticas de Desplazamiento.



$$\sum F_y = 0 \Rightarrow \theta_{CD} = 4500 / EI_0$$



Viga 1:

$$\sum F_x = 0 \Rightarrow \theta_{CD} = 4500 / EI_o$$

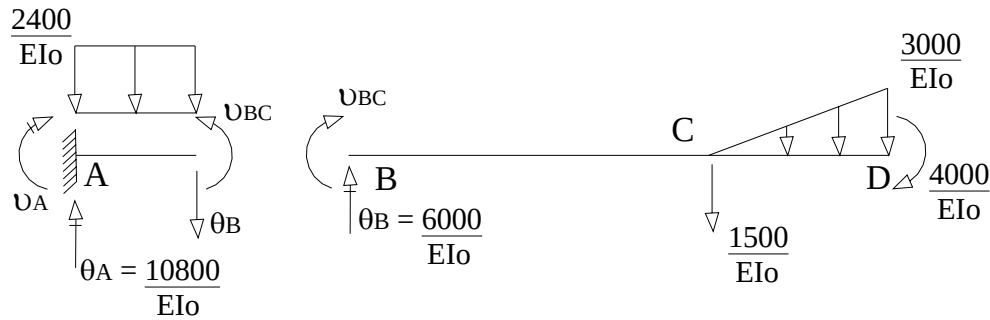
$$\begin{aligned} \sum M = 0 &\Rightarrow \bar{\mu}_C + \frac{2250}{EI_o} \times \frac{4}{2} \times \frac{2}{3} \times 4 = 0 \\ &\Rightarrow \bar{\mu}_C = -12000 / EI_o \end{aligned}$$

Viga 2:

$$\sum F_y = 0 \Rightarrow \theta_B = 6000 / EI_o$$

$$\begin{aligned} \sum M = 0 &\Rightarrow v_{BE} + \frac{2000}{EI_o} \times 3 + \frac{4000}{3EI_o} \times \frac{3^2}{2} = 0 \\ &\Rightarrow v_{BE} = 12000 / EI_o \end{aligned}$$

Viga 3:



$$\sum F_y = 0 \Rightarrow \theta_A = 10800 / EI_o$$

$$\begin{aligned} \sum M_B^{BCD} = 0 &\Rightarrow v_{BC} + \frac{40000}{EI_o} + \frac{1500}{EI_o} \times 6 + \frac{3000}{EI_o} \times \frac{3}{2} \times 8 = 0 \\ &\Rightarrow v_{BC} = -85000 / EI_o \end{aligned}$$

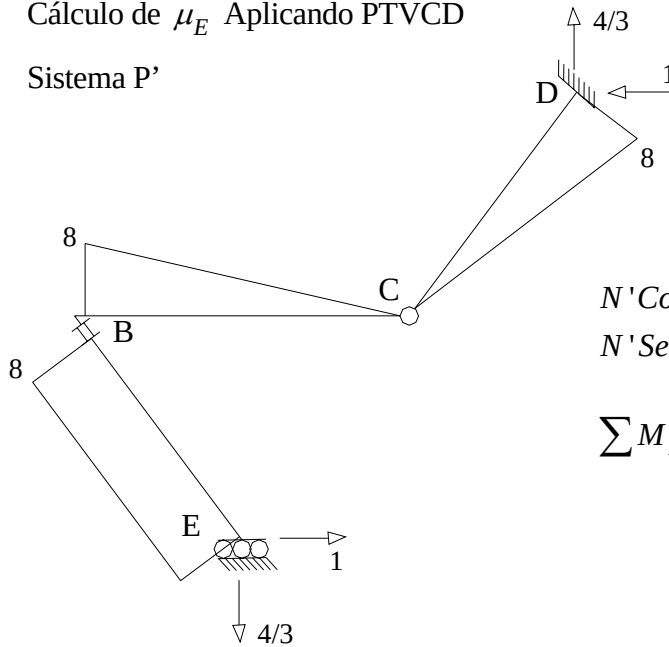
$$\begin{aligned} \sum M_B^{AB} = 0 &\Rightarrow \frac{10800}{EI_o} \times 2 + v_A + \frac{85000}{EI_o} - \frac{2400}{EI_o} \times 2 \times 1 = 0 \\ &\Rightarrow v_A = -101800 / EI_o \end{aligned}$$

Viga 4:

$$\begin{aligned} \sum F_x = 0 &\Rightarrow \theta_B - \frac{1000}{EI_o} \times 4 - \frac{2000}{EI_o} = 0 \\ &\Rightarrow \theta_B = 6000 / EI_o \end{aligned}$$

Cálculo de μ_E Aplicando PTVCD

Sistema P'



$$N' \cos \phi = 1 // \cos \phi = 3/5 \Rightarrow N' = 5/3$$

$$N' \sin \phi = A'_v \Rightarrow A'_v = 4/3$$

$$\sum M_B = 0 \Rightarrow M'_E - 1 \times 4 - \frac{4}{3} \times 3 = 0$$

$$\Rightarrow M'_E = 8$$

$$\Rightarrow M'_{DC} = 8$$

$$\begin{aligned} T_R P' &= -2 \times 10^{-3} \times 8 - 4 \times 10^{-2} \times \frac{4}{3} \\ &= -\frac{16000}{EI_0} - \frac{53333}{EI_0} \\ &= -\frac{69333}{EI_0} \end{aligned}$$

$$\begin{aligned} T_i &= \frac{2400 \times 8 \times 5}{3EI_0} + \frac{1}{3} \times \frac{3600 \times 8 \times 5}{2EI_0} \\ &= \frac{56000}{EI_0} \end{aligned}$$

De donde:

$$1 \times \mu_E - \frac{69333}{EI_0} = \frac{56000}{EI_0}$$

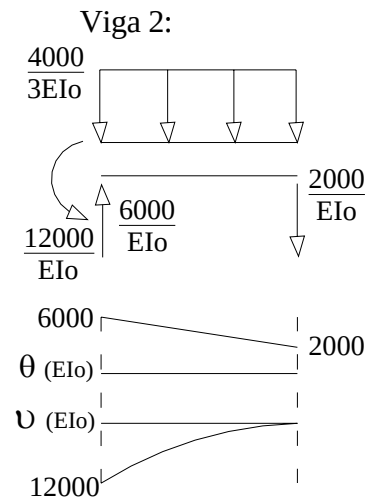
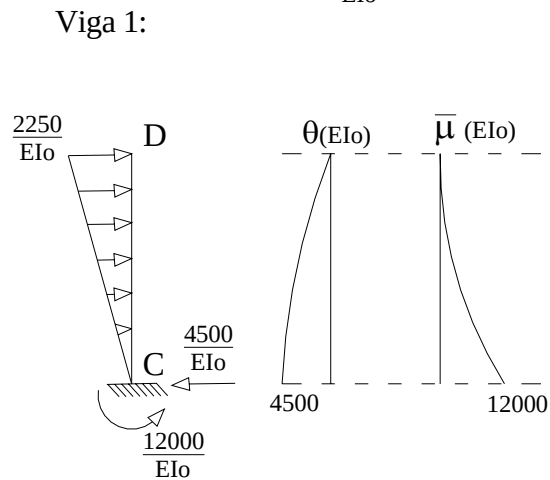
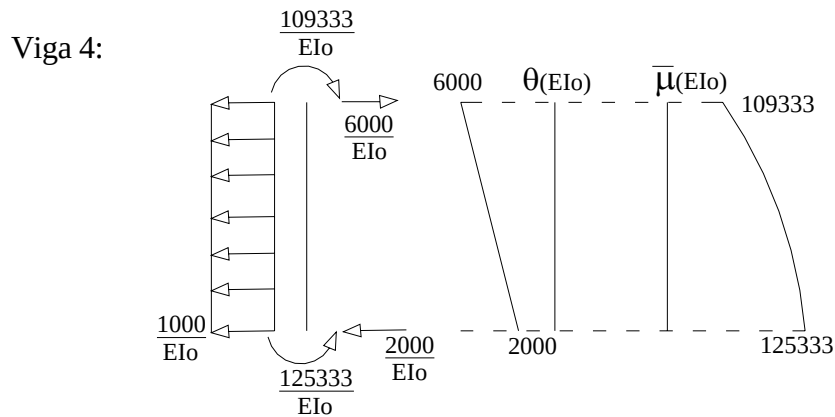
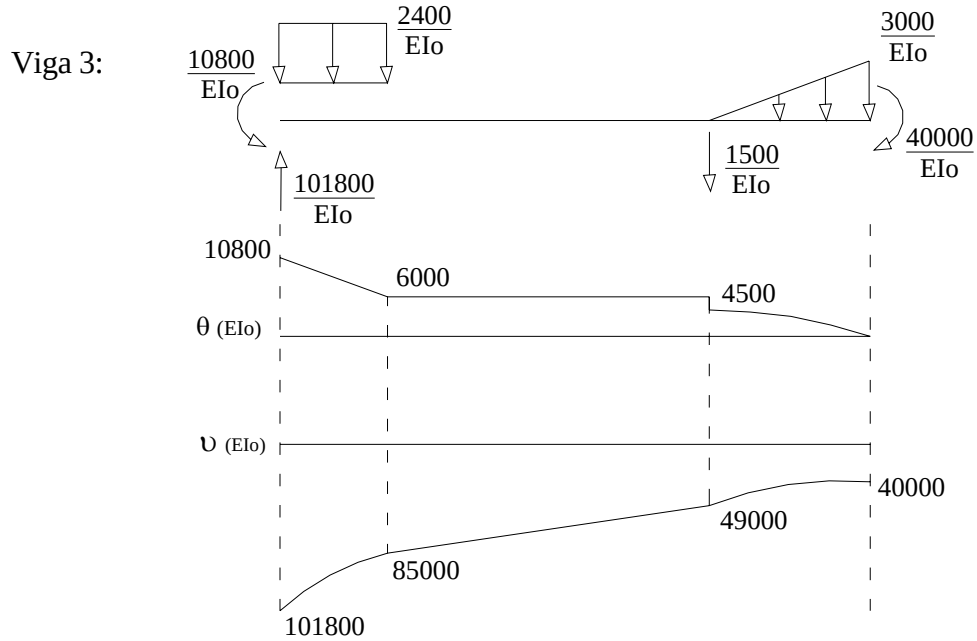
$$\mu_E = \frac{125333}{EI_0}$$

$$\mu_E = -\frac{125333}{EI_0}$$

Continuando con la Viga 4 Tenemos:

$$\begin{aligned} \sum M &= 0 \Rightarrow \mu_{BE} + \frac{125333}{EI_0} - \frac{2000}{EI_0} \times 4 - \frac{1000}{EI_0} \times \frac{4^2}{2} \\ &\Rightarrow \mu_{BE} = -\frac{109333}{EI_0} \end{aligned}$$

Diagramas de las Vigas Conjugadas:



2.- Para la Estructura Indicada, Determinar:

a.- $S = v_D + 4\theta_B$, Aplicando Trabajo Virtual

b.- Elásticas de Desplazamientos.

$$\theta_E = -5 \times 10^{-3} \text{ rad}$$

$$EI_o = 10^6 \text{ Kg.m}^2$$

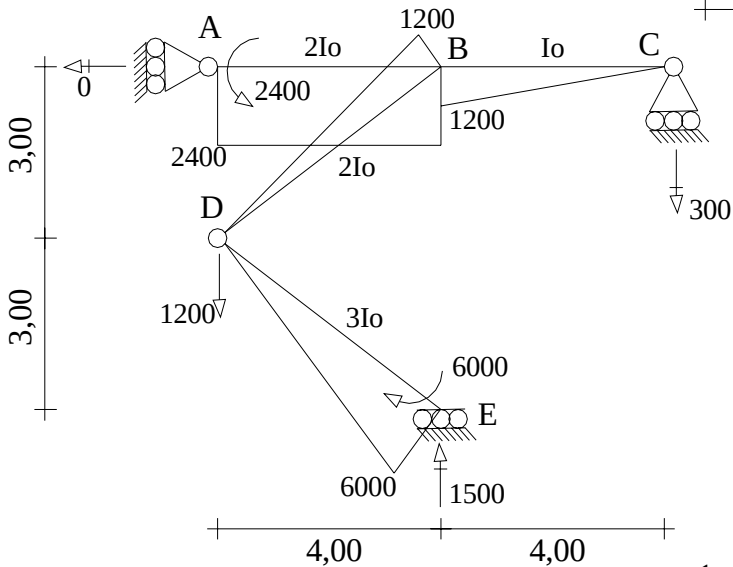
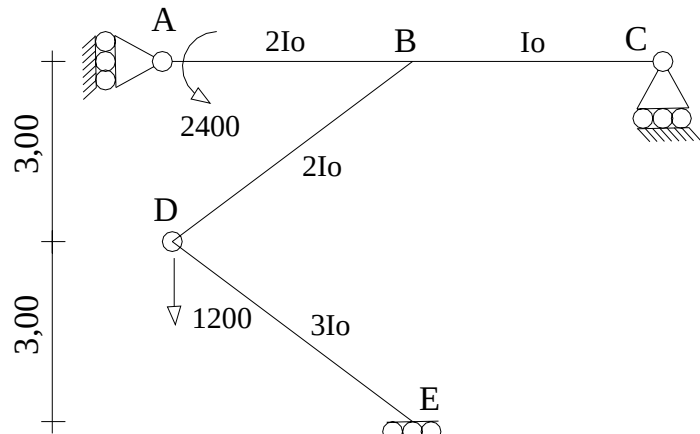
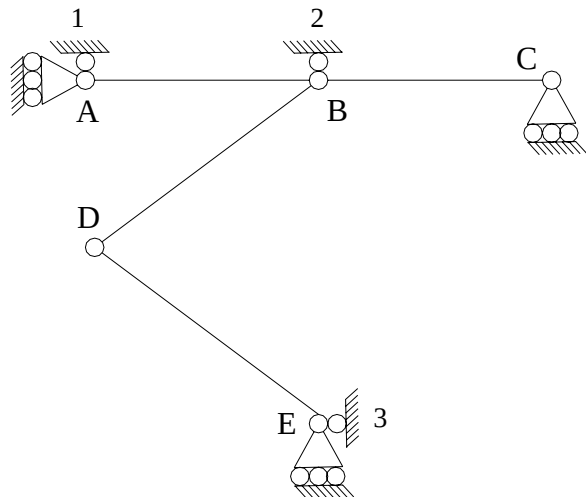
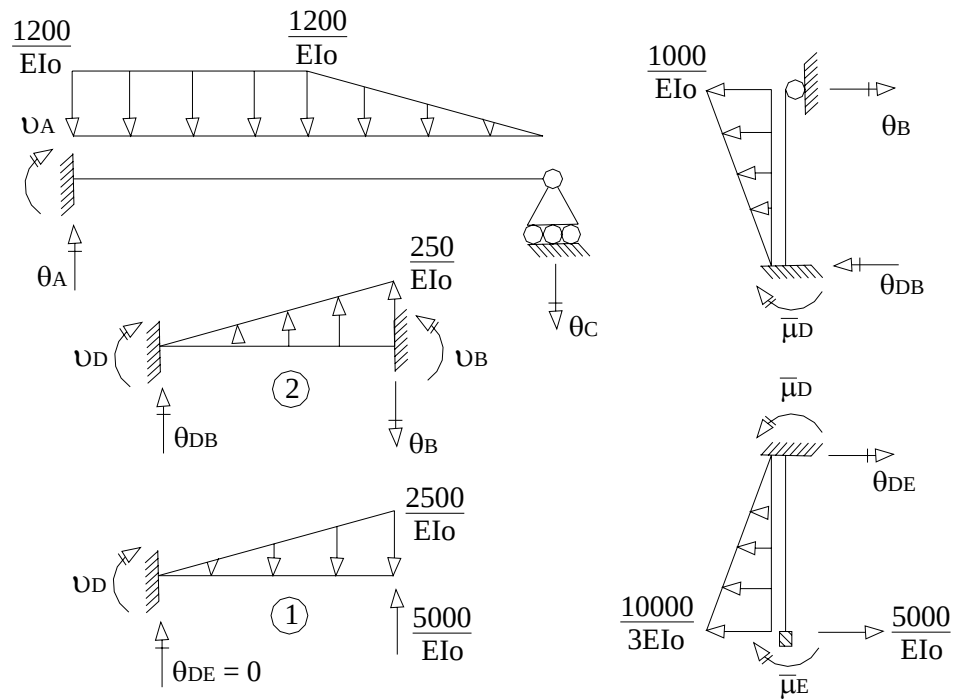


Imagen Cinemática:



Vigas Conjugadas:



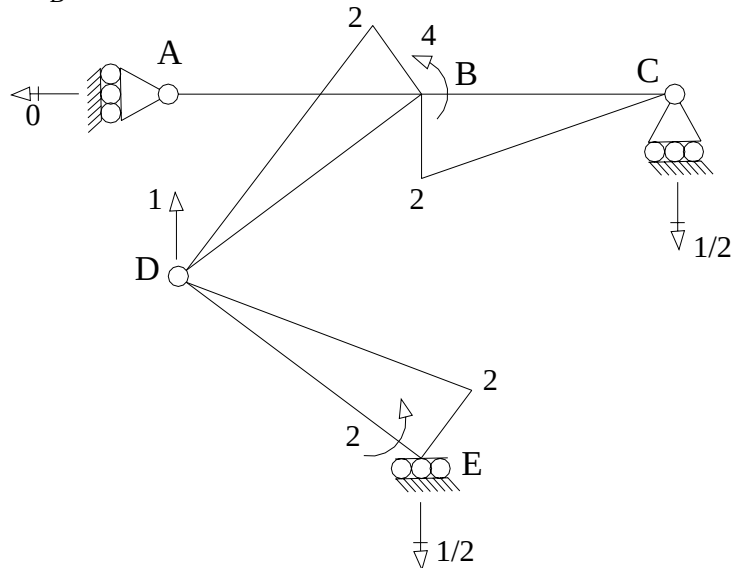
$$\theta_{DE} = 0$$

$$v_D = \frac{5000}{EI_0} \times 4 - \frac{5000}{EI_0} \times \frac{2}{3} \times 4$$

$$v_D = \frac{6667}{EI_0}$$

Cálculo de $S = v_D + 4\theta_B$

Sistema P'



$$S = \frac{8533}{EI_0}$$

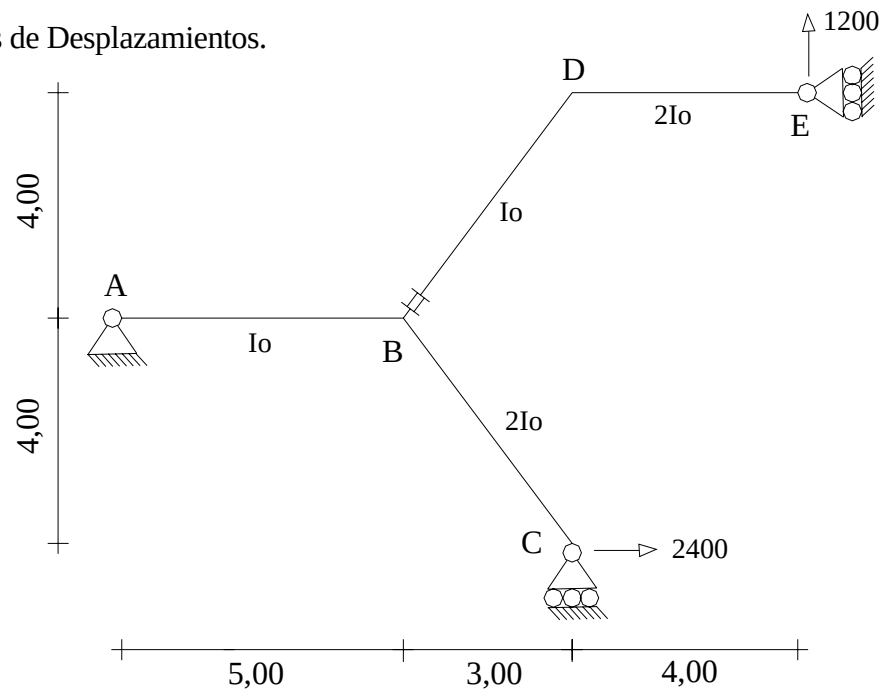
$$\text{Si } \nu_D = \frac{6667}{EI_0} \Rightarrow \theta_B = \left(\frac{8533}{EI_0} - \frac{6667}{EI_0} \right) \frac{1}{4} = \frac{466,5}{EI_0}$$

$$\theta_B = \frac{467}{EI_0}$$

3.- Determinar

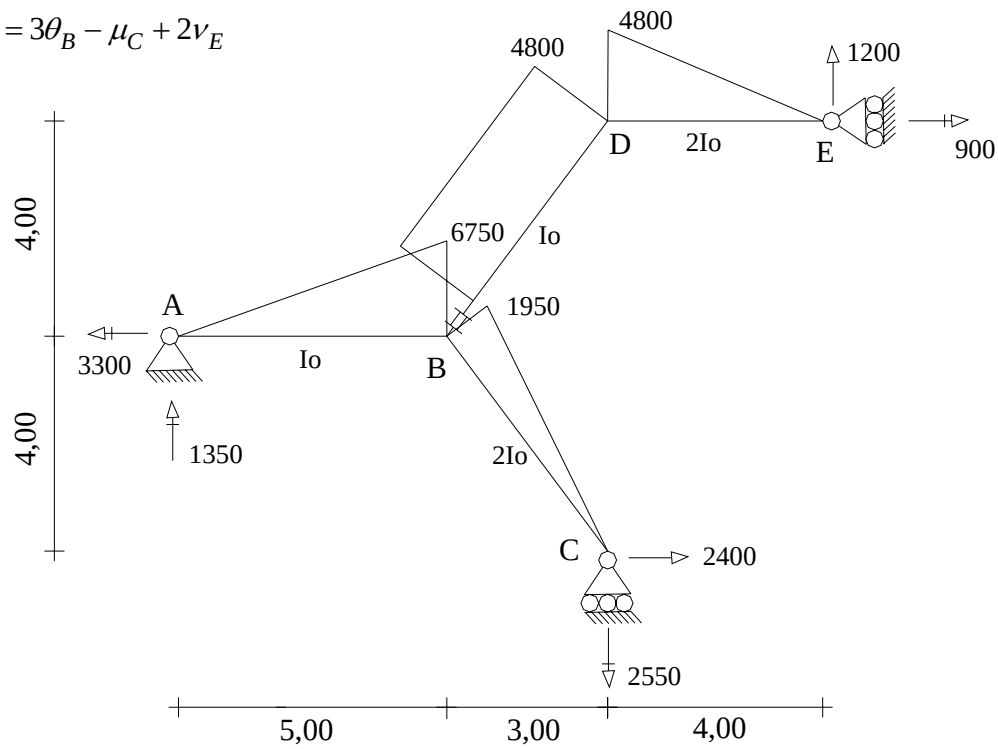
a.- $S = 3\theta_B - \mu_C + 2v_E$, Aplicando PTVCD.

b.- Elásticas de Desplazamientos.

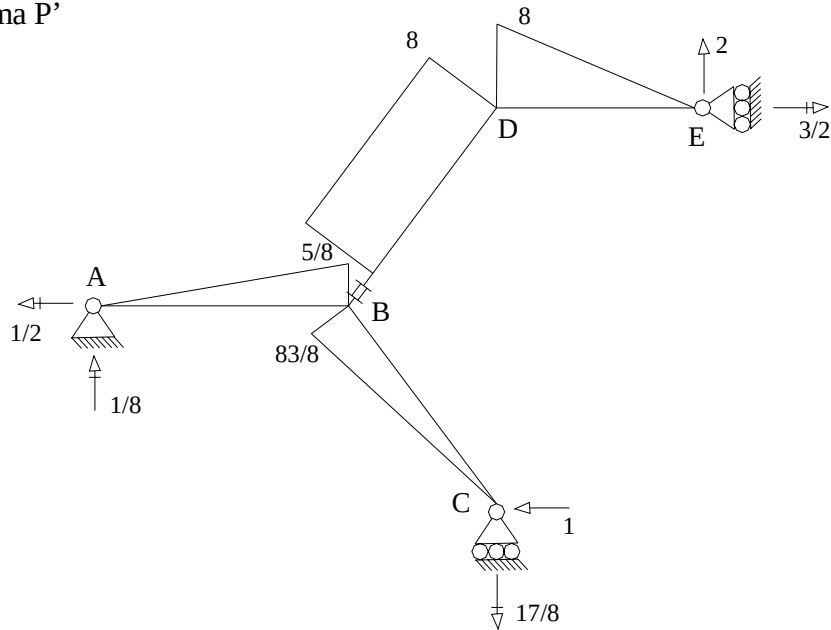


Sistema P

$$S = 3\theta_B - \mu_C + 2v_E$$



Sistema P'

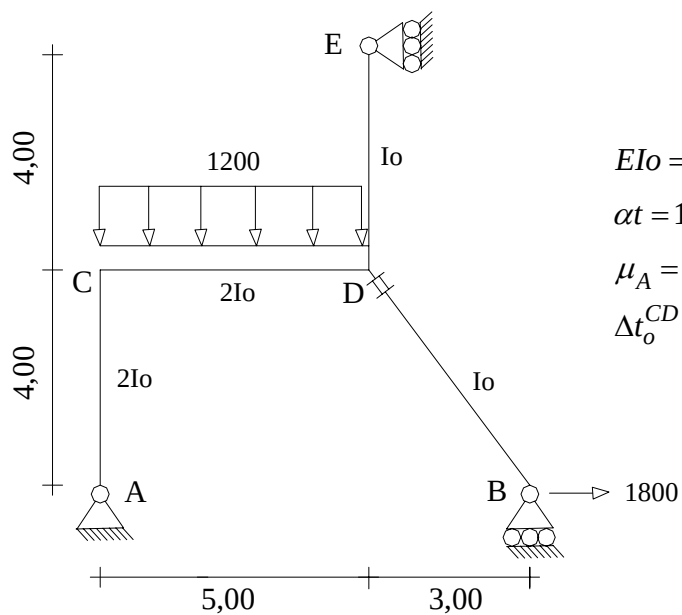


$$S = \frac{1}{3}x \frac{(5/8)x6750x5}{EI_o} - \frac{1}{3}x \frac{1950x(83/8)x5}{2EI_o} + \frac{8x4800x5}{EI_o} + \frac{1}{3}x \frac{4800x8x4}{2EI_o} - \left(\frac{-1}{8}x2x10^{-2}x \frac{10^6}{EI_o}\right)$$

$$S = \frac{7031,25}{EI_o} - \frac{16859,37}{EI_o} + \frac{192000}{EI_o} + \frac{25600}{EI_o} + \frac{2500}{EI_o}$$

$$S = \frac{210271,88}{EI_o}$$

4.- Calcular: $S = 2\psi_{DB} - 3\nu_E - \frac{\theta_C}{4}$



$$EI_o = 10^6 \text{ Kg.m}^2$$

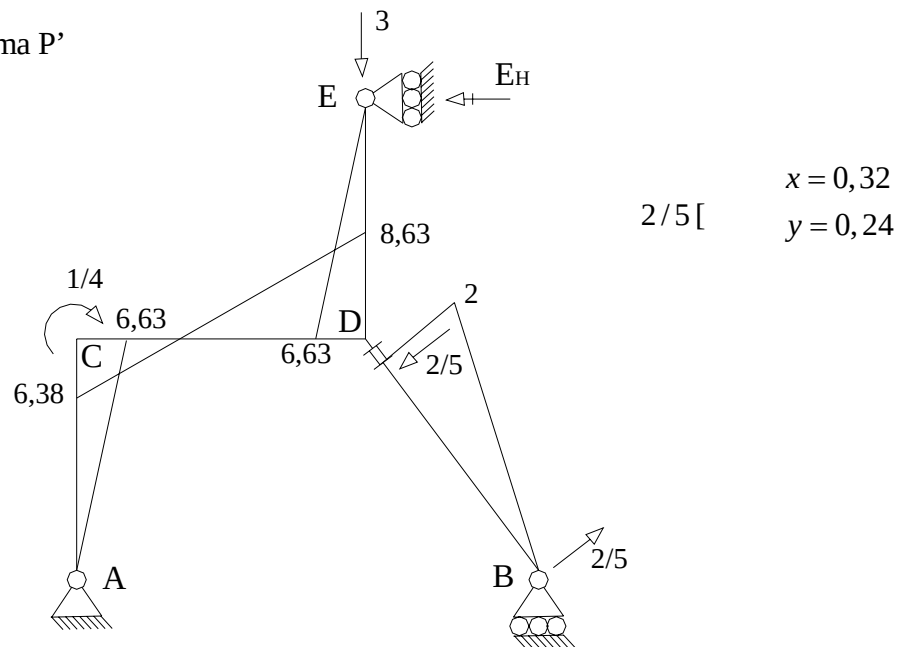
$$\alpha t = 1 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$$

$$\mu_A = -3 \times 10^{-2} m$$

$$\Delta t_o^{CD} = 10^\circ\text{C}$$

Solución:

Sistema P'



$$\sum M_B^{DB} = 0 \Rightarrow 2 = M'_{DB} \curvearrowright$$

$$\sum M_D^{DB} = 0 \Rightarrow B'_V = 0 \Rightarrow N'_{DB} = 0$$

$$\sum M_A = 0 \Rightarrow 2 + E'_H \times 8 = 3 \times 5 + 1/4 \Rightarrow E'_H = 53/32 = 1,66$$

$$\sum F_y = 0 \Rightarrow A'_V = 3 \uparrow \quad // \quad \sum F_x = 0 \Rightarrow A'_H = 1,66 \rightarrow$$

$$N'_{CD} = -1,66 \quad // \quad \Omega_{N'}^{CD} = -8,30$$

$$T_R.P' = -3 \times 10^{-2} \times 1,66 = -4,98 \times 10^{-2}$$

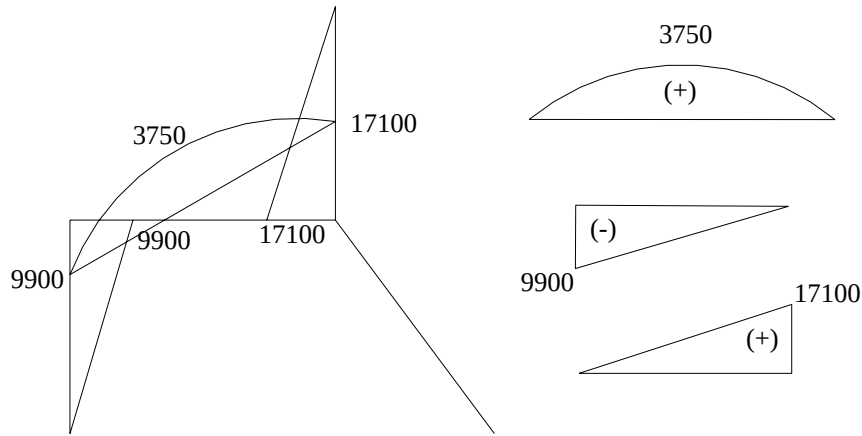
Sistema P

$$\sum M_B^{DB} = 0 \Rightarrow M_{DB} = 0$$

$$\sum M_D^{DB} = 0 \Rightarrow 1800 \times 4 = B_V (\downarrow) \times 3 \Rightarrow B_V = 2400 \downarrow$$

$$\sum M_A = 0 \Rightarrow 2400 \times 8 + 1200 \times 5^2 / 2 = E_H (\leftarrow) \times 8 \Rightarrow E_H = 4275 \leftarrow$$

$$\sum F_y = 0 \Rightarrow A_V = 8400 \uparrow \quad // \quad \sum F_x = 0 \Rightarrow A_H = 2475 \rightarrow$$



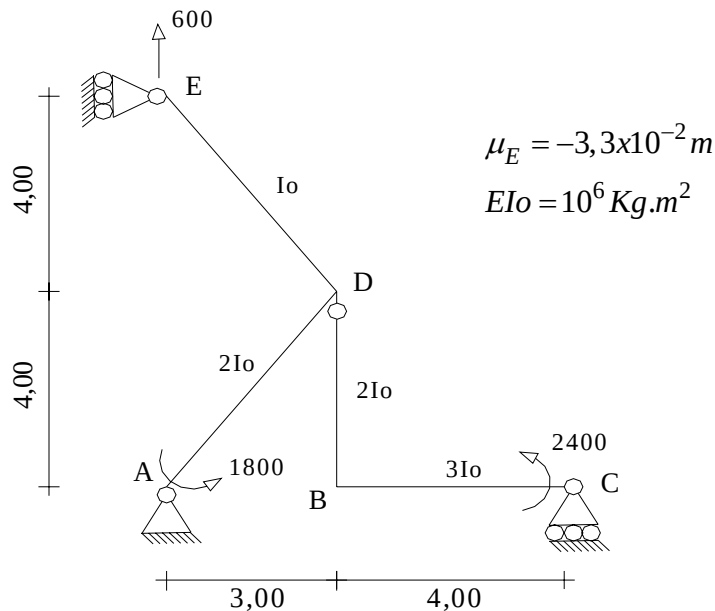
$$T_i^{AC} = \frac{1}{3} \times 9900 \times \frac{6,63 \times 4}{2EI_o} = \frac{43758}{EI_o}$$

$$T_i^{CD} = \frac{1}{3} \times 6,38 \times \frac{3750 \times 5}{2EI_o} + \frac{1}{3} \times 6,38 \times \frac{9900 \times 5}{2EI_o} - \frac{1}{6} \times 6,38 \times \frac{17100 \times 5}{2EI_o} + \frac{1}{3} \times 8,63 \times \frac{3750 \times 5}{2EI_o} + \frac{1}{3} \times 8,63 \times \frac{17100 \times 5}{2EI_o} - \frac{1}{6} \times 8,63 \times \frac{9900 \times 5}{2EI_o} + 10^{-5} \times 10(-8,30) \times \frac{10^6}{EI_o} = \frac{100757,5}{EI_o}$$

$$T_i^{DE} = \frac{1}{3} \times 6,63 \times \frac{17100 \times 4}{EI_o} = \frac{151164}{EI_o}; \quad T_i^{DB} = 0$$

$$T_i^{Total} = \frac{295679,50}{EI_o} \Rightarrow S = \frac{295679,50}{EI_o} - \left(-\frac{49800}{EI_o}\right) = \frac{345479,5}{EI_o}$$

5.- Para la estructura indicada determinar Elásticas de Desplazamientos.



Solución:

a) Cálculo de las Reacciones:

$$\sum M_D = 0 \Rightarrow C_V = 800 \downarrow$$

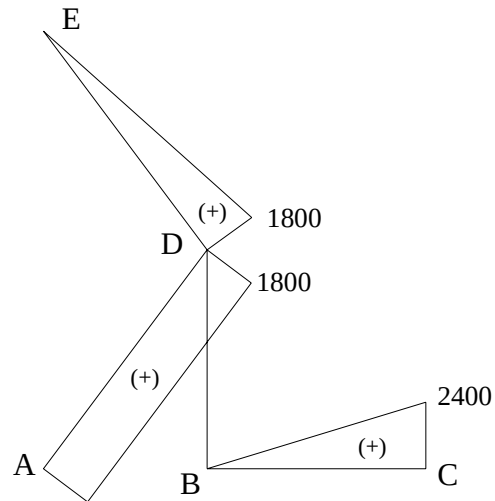
$$\sum M_A = 0 \Rightarrow E_H (-) \times 8 + 600 \times 7 = 2400 + 1800 \Rightarrow E_H = 0$$

$$\sum F_y = 0 \Rightarrow A_V = 0$$

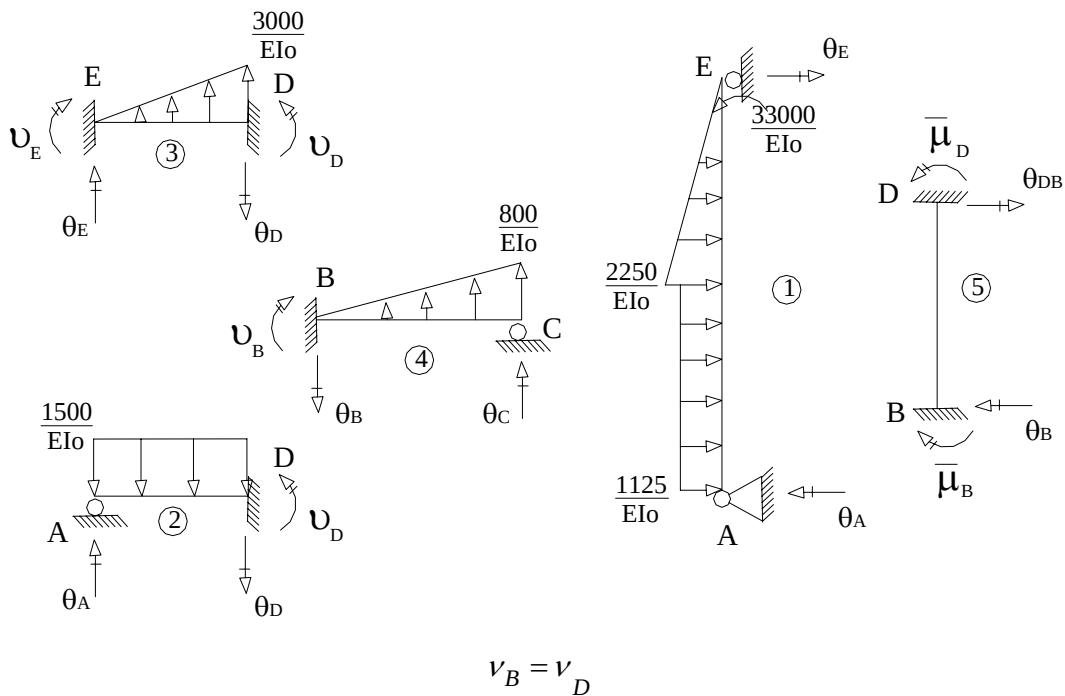
$$\sum F_x = 0 \Rightarrow A_H = 0$$

b) Diagramas de Momento:

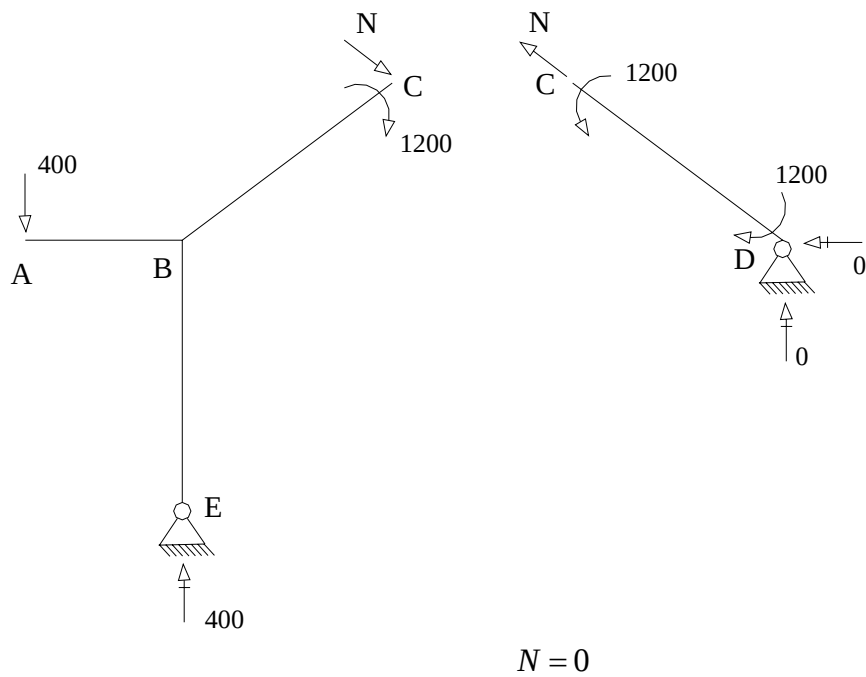
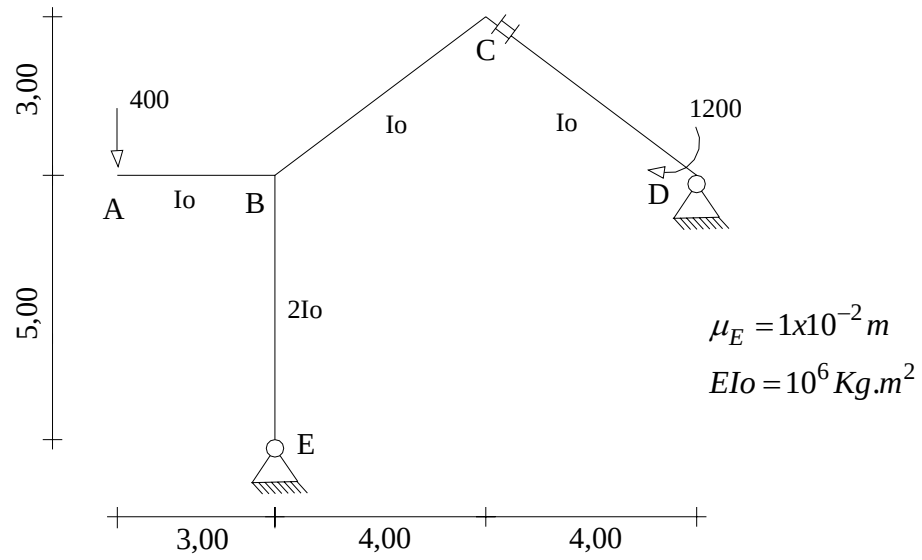
$$\bar{\mu}_E = 3,3 \times 10^{-2} \times \frac{10^6}{EI_0} = \frac{33000}{EI_0}$$

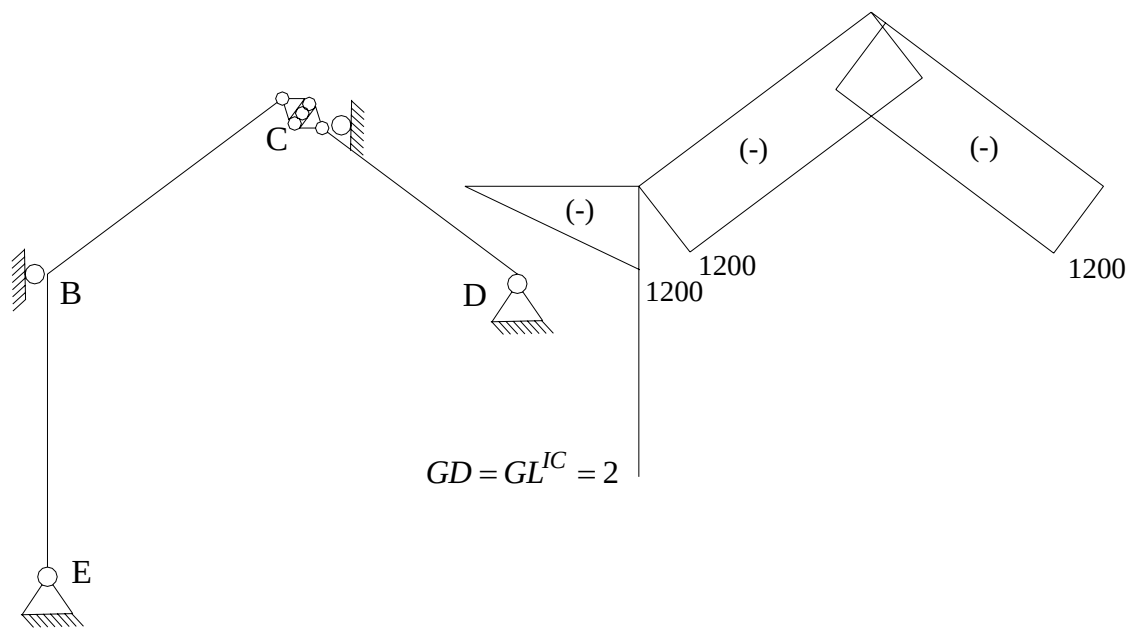


c) Vigas Conjugadas

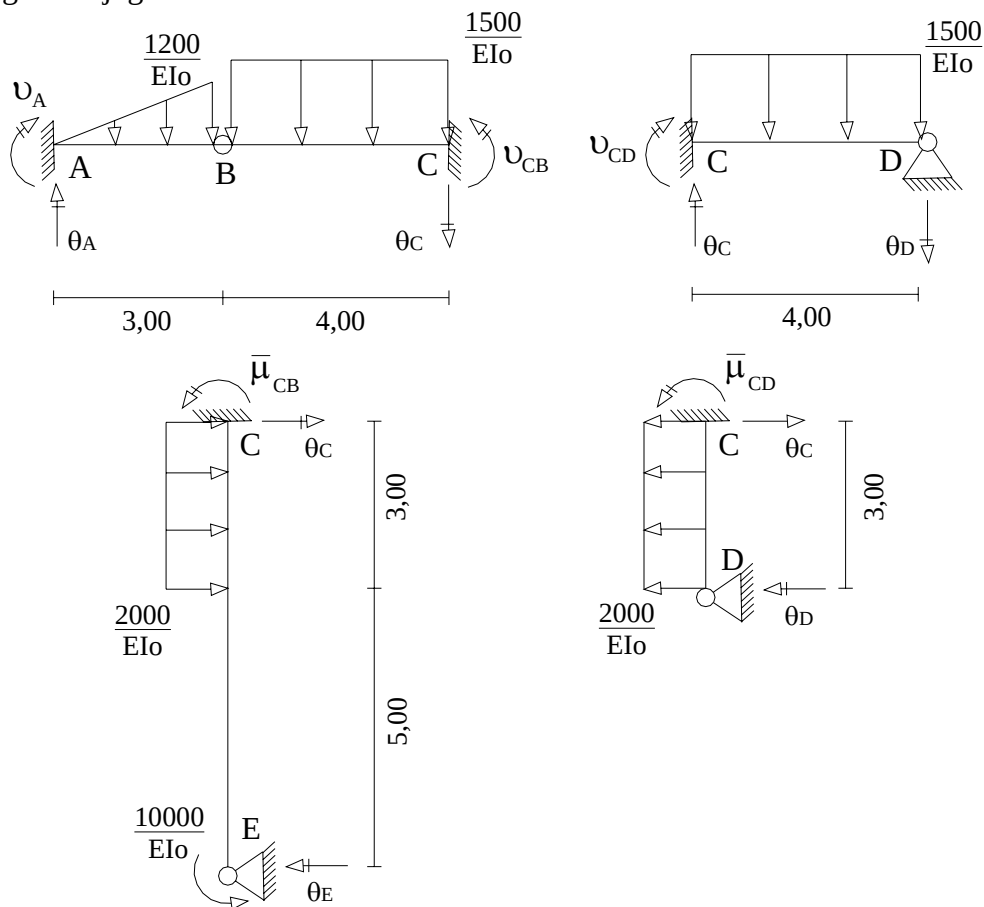


6.- Determinar Elásticas de Desplazamientos.





Vigas Conjugadas:



7.- Calcular:

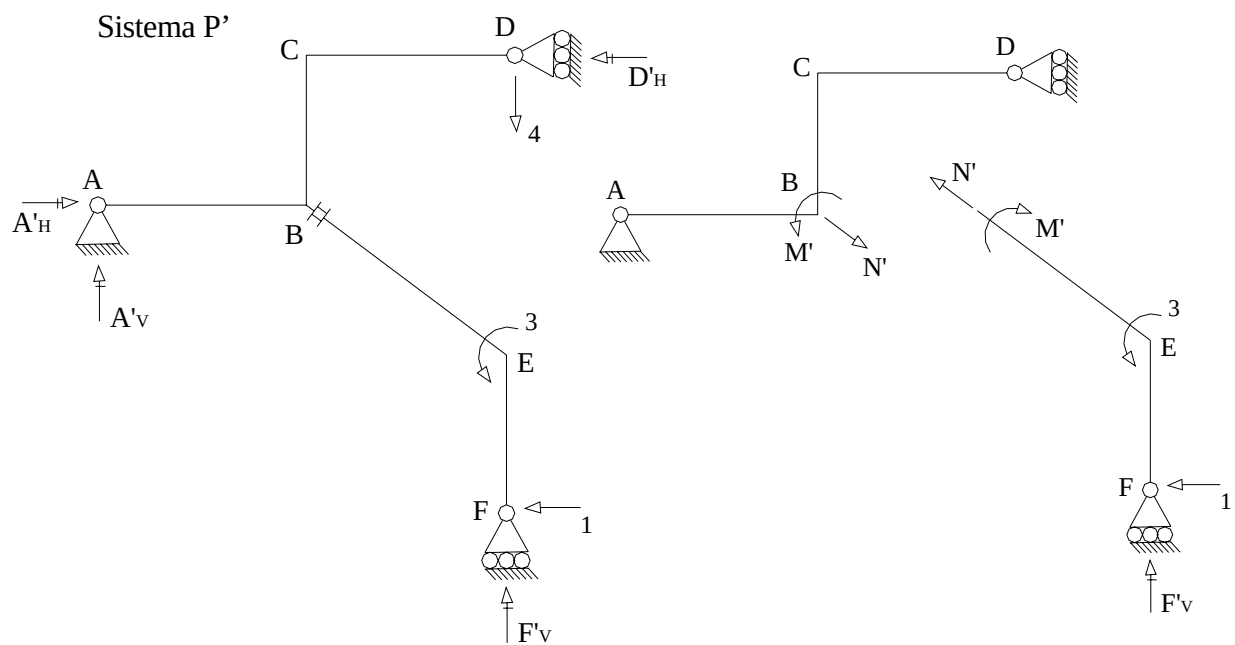
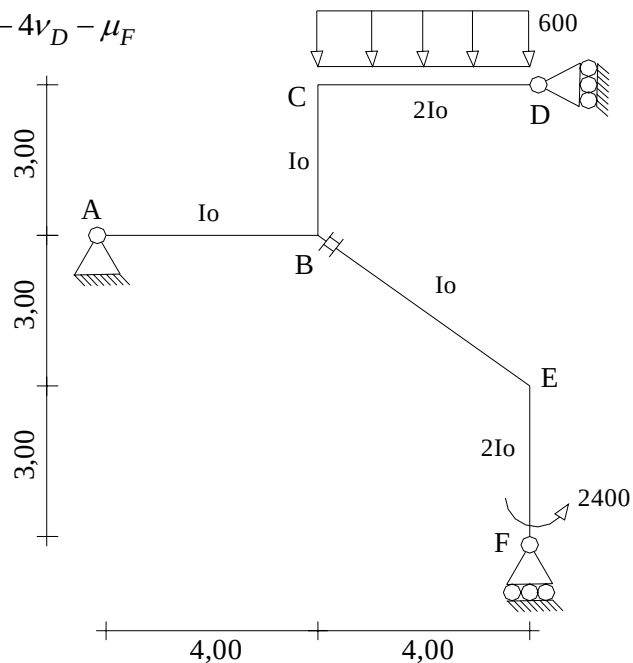
$$S = 3\theta_E - 4v_D - \mu_F$$

$$v_F = -2 \times 10^{-2} m$$

$$\Delta t_o = 15^\circ C$$

$$\alpha t = 1 \times 10^{-5} ^\circ C^{-1}$$

$$EI_o = 10^6 Kg.m^2$$



$$\sum M_E^{BEF} = 0 \Rightarrow M'_B = 0$$

$$\sum M_B^{BEF} = 0 \Rightarrow F'_V \times 4 + 3 = 1 \times 6 \Rightarrow F'_V = 3/4$$

$$\sum M_A = 0 \Rightarrow D'_H \times 3 + 3 + 3/4 \times 8 = 4 \times 8 + 1 \times 6 \Rightarrow D'_H = 29/3$$

$$\sum F_H = 0 \Rightarrow A'_H = 1 + D'_H \Rightarrow A'_H = 32/3$$

$$\sum F_y = 0 \Rightarrow A'_V + F'_V = 4 \Rightarrow A'_V = \frac{16}{4} - \frac{3}{4} \Rightarrow A'_V = \frac{13}{4}$$

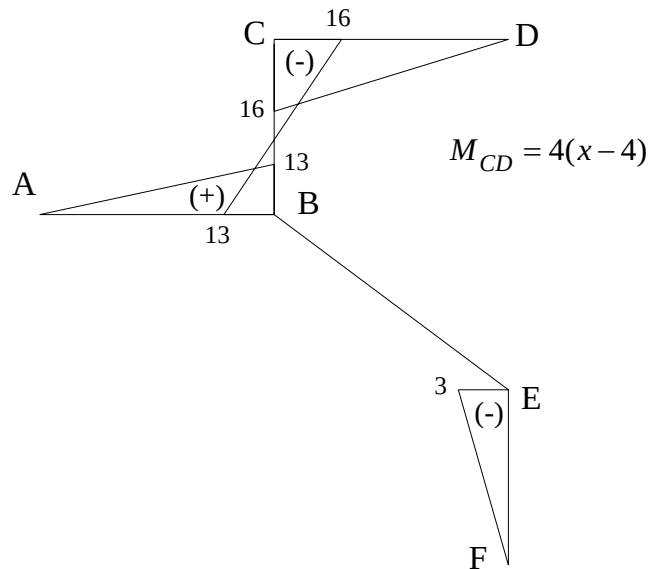
Para BE: $\bar{X} = 1$ y $\bar{y} = -3/4$

$$N' = -(1)x\frac{4}{5} - \left(\frac{-3}{4}\right)\left(\frac{-3}{5}\right)$$

$$N' = \frac{-16}{20} - \frac{9}{20} = \frac{-25}{20} = \frac{-5}{4}$$

$$\Omega_{N'}^{BE} = \frac{-5}{4} \times 5 = \frac{-25}{4}$$

Diagramas de Momento:



Solución del Sistema P:

$$\sum M_E^{BEF} = 0 \Rightarrow M_{BE} = 2400$$

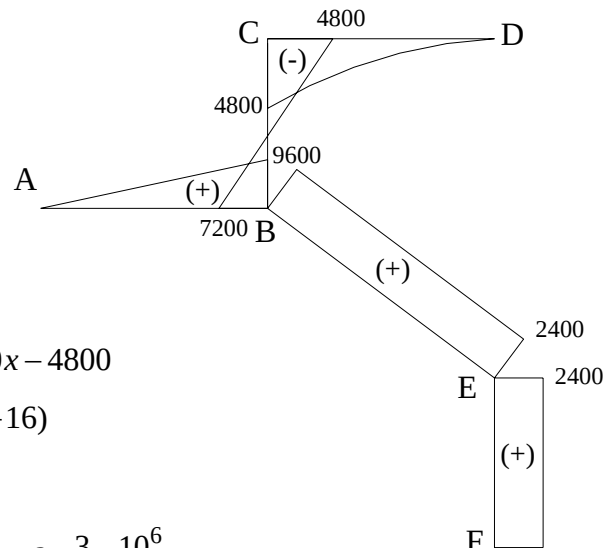
$$\sum M_B^{BEF} = 0 \Rightarrow F_V = 0$$

$$\sum M_A = 0 \Rightarrow D_H(\leftarrow) \times 3 + 2400 = 600 \times 4 \times 6 \Rightarrow D_H = 4000 \leftarrow$$

$$\sum F_H = 0 \Rightarrow A_H = 4000 \rightarrow$$

$$\sum F_y = 0 \Rightarrow A_V = 2400 \uparrow$$

Diagramas de Momento:



$$M_{CD} = -300x^2 + 2400x - 4800$$

$$M_{CD} = -300(x^2 - 8x + 16)$$

$$\text{Trabajo Externo} = S + \left(-2 \times 10^{-2} \times \frac{3}{4} \times \frac{10^6}{EI_0}\right)$$

$$\text{Trabajo Externo} = S - 15000 / EI_0$$

Trabajo Interno:

$$T_i^{AB} = \frac{1}{3} \times \frac{9600 \times 13 \times 4}{EI_o} = \frac{166400}{EI_o}$$

$$T_i^{BC} = \frac{1}{3} \times \frac{13 \times 7200 \times 3}{EI_o} + \frac{1}{3} \times \frac{16 \times 4800 \times 3}{EI_o} - \frac{1}{6} \times \frac{13 \times 4800 \times 3}{EI_o} - \frac{1}{6} \times \frac{16 \times 7200 \times 3}{EI_o}$$

$$T_i^{BC} = \frac{81600}{EI_o}$$

$$T_i^{BE} = 10^{-5} \times 15 \times \frac{25}{4} \times \frac{10^6}{EI_o} = \frac{937,5}{EI_o}$$

$$T_i^{EF} = \frac{-1}{2} \times \frac{3 \times 2400 \times 3}{2EI_o} = \frac{-5400}{EI_o}$$

$$T_i^{CD} = \int_0^4 \frac{4(x-4)}{2EI_o} [-300(x^2 - 8x + 16)] dx$$

$$T_i^{CD} = \frac{-600}{EI_o} \int_0^4 (x-4)[-300(x^2 - 8x + 16)] dx$$

$$T_i^{CD} = \frac{-600}{EI_o} (-64) = \frac{38400}{EI_o}$$

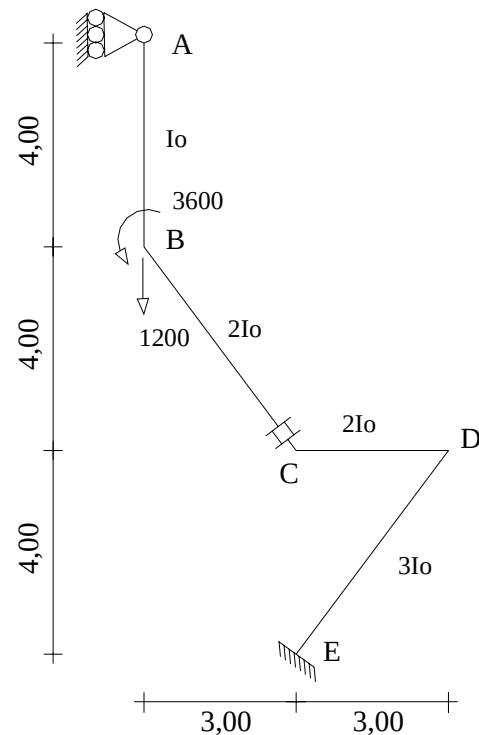
Luego: $T_i^{Total} = 280062,50 / EI_o$

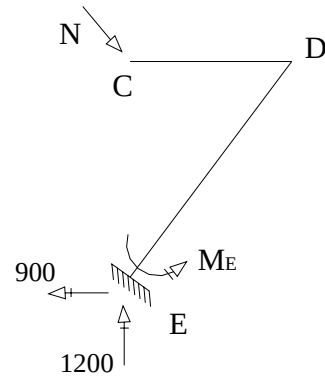
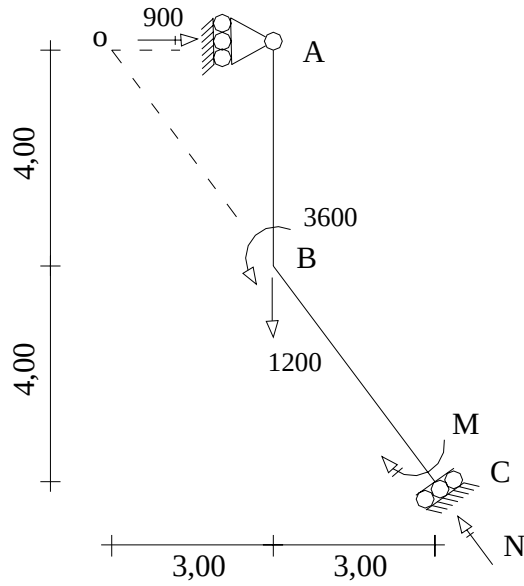
De donde: $S = 295062,50 / EI_o$

8.- Determinar Elásticas de Desplazamiento.

$$\theta_E = 3 \times 10^{-3} \text{ rad}$$

$$EI_o = 10^6 \text{ Kg.m}^2$$





$$\sum M_o = 0 \Rightarrow M = 0$$

$$\begin{aligned} \sum M_C = 0 &\Rightarrow 1200 \times 3 + 3600 = A_H \times 8 \\ &\Rightarrow 7200 / 8 = 900 \end{aligned}$$

$$\sum F_H = 0 \Rightarrow E_H = 900 \text{ (para toda la estructura)}$$

$$\sum F_V = 0 \Rightarrow E_V = 1200 \uparrow$$

$$\begin{aligned} \sum M_C^{CDE} = 0 &\Rightarrow 900 \times 4 = M_E \\ &\Rightarrow M_E = 3600 \end{aligned}$$

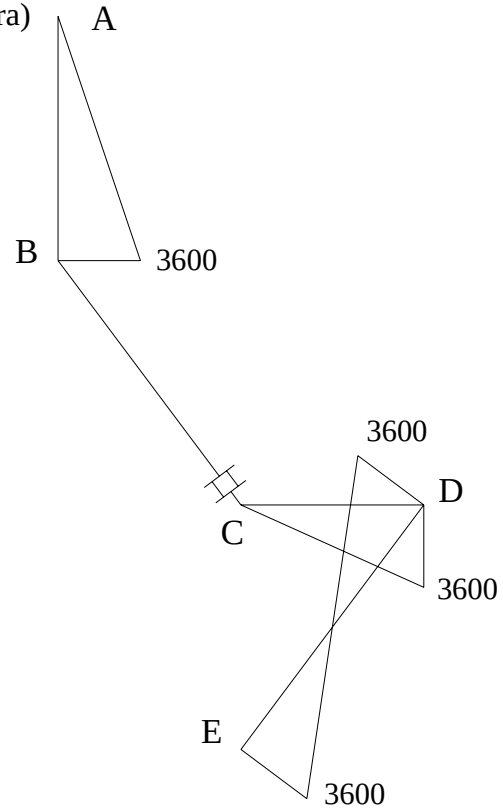
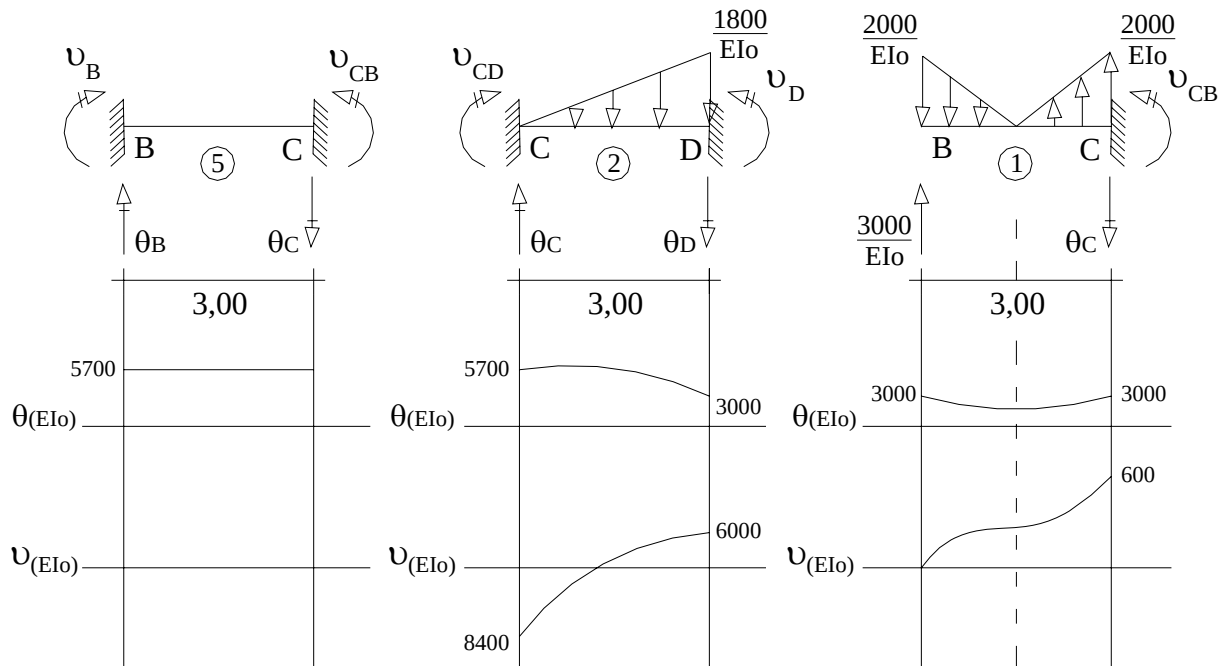


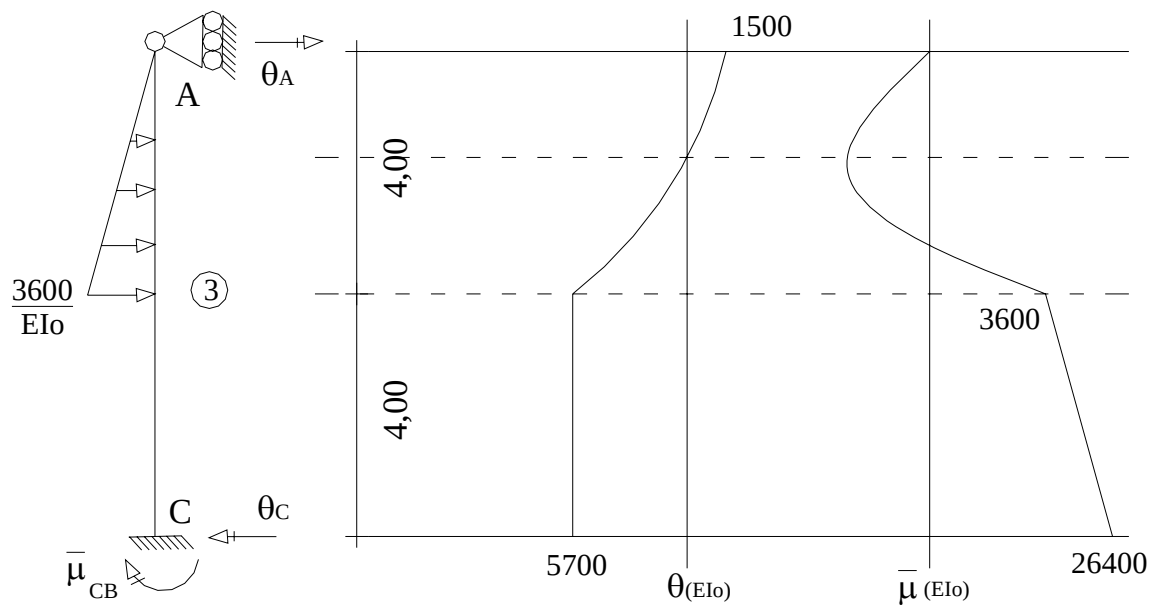
Diagrama de Momento:

Vigas Conjugadas:

a.- Para Elásticas de Desplazamientos Verticales:



b.- Para Elásticas de Desplazamientos Horizontales:



Solución Viga 1:

$$\sum F_y = 0 \Rightarrow \theta_D = 3000 / EI_o$$

$$\begin{aligned} \sum M_D = 0 \Rightarrow \frac{3000}{EI_o} \times 3 + \frac{2000}{EI_o} \left(\frac{3}{2}\right) \frac{1}{2} \left(\frac{1}{3}\right) \frac{3}{2} &= v_D + \frac{2000}{EI_o} \left(\frac{3}{2}\right) \frac{1}{2} \left(\frac{5}{2}\right) \\ \Rightarrow v_D &= 6000 / EI_o \end{aligned}$$

Solución Viga 2: Tenemos $v_D = 6000 / EI_o$ // $\theta_D = 3000 / EI_o$

$$\begin{aligned} \sum F_y = 0 \Rightarrow \theta_C &= \frac{3000}{EI_o} + \frac{1800}{EI_o} \times \frac{3}{2} \\ \Rightarrow \theta_C &= 5700 / EI_o \end{aligned}$$

$$\begin{aligned} \sum M_C = 0 \Rightarrow v_{CD} + \frac{3000}{EI_o} \times 3 + \frac{1800}{EI_o} \times \frac{3}{2} \times 2 &= v_D \\ \Rightarrow v_{CD} &= -8400 / EI_o \end{aligned}$$

Solución Viga 3: Tenemos $\theta_C = 5700 / EI_o$

$$\begin{aligned} \sum F_x = 0 \Rightarrow \theta_C &= \frac{5700}{EI_o} = \theta_A + \frac{3600}{EI_o} \times \frac{4}{2} \\ \Rightarrow \theta_A &= -1500 / EI_o \end{aligned}$$

$$\begin{aligned} \sum M_C = 0 \Rightarrow \bar{\mu}_{CB} + \frac{3600}{EI_o} \times \frac{4}{2} \left(\frac{16}{3}\right) &= \frac{1500}{EI_o} \times 8 \\ \Rightarrow \bar{\mu}_{CB} &= -26400 / EI_o \end{aligned}$$

Solución Viga 4: Tenemos $\theta_D = 3000 / EI_o$

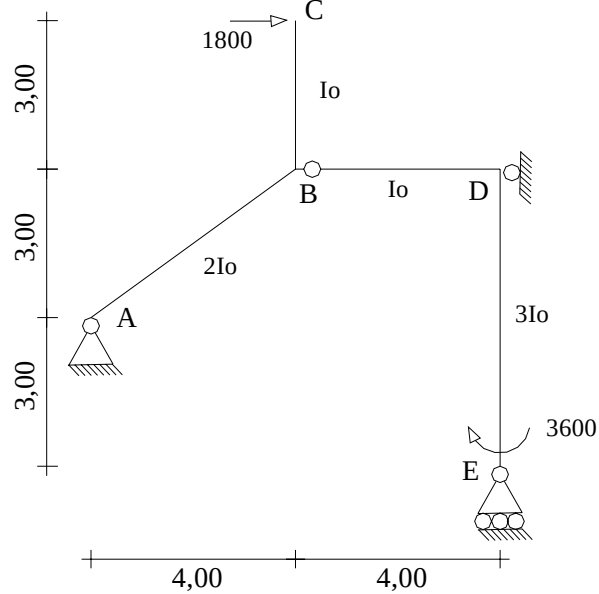
$$\begin{aligned} \sum M_D = 0 \Rightarrow \bar{\mu}_D + \frac{1500}{EI_o} \times \frac{2}{2} \times \frac{10}{3} &= \frac{3000}{EI_o} \times 4 + \frac{1500}{EI_o} \times \frac{2}{2} \times \frac{2}{3} \\ \Rightarrow \bar{\mu}_D &= 8000 / EI_o \end{aligned}$$

Solución Viga 5:

$$\theta_C = \theta_B = 5700 / EI_o$$

v_B o v_{CB} se calcularán aplicando Trabajo Virtual.

9.- Determinar Elásticas de Desplazamientos.

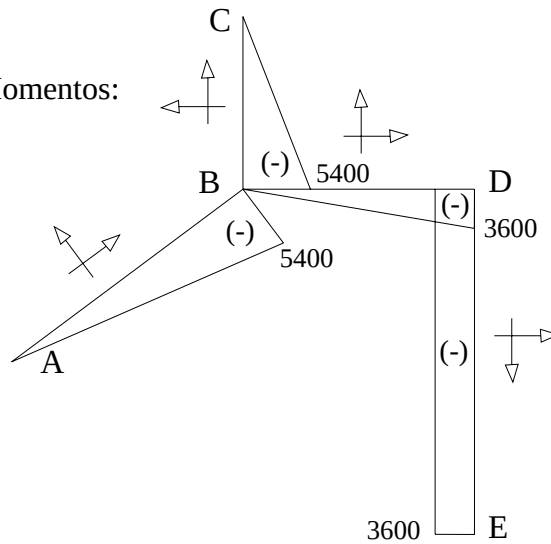


$$v_A = -4 \times 10^{-2} m$$

$$EI_0 = 10^7 Kg.m$$

Solución:

Diagramas de Momentos:



Movimiento de Apoyos:

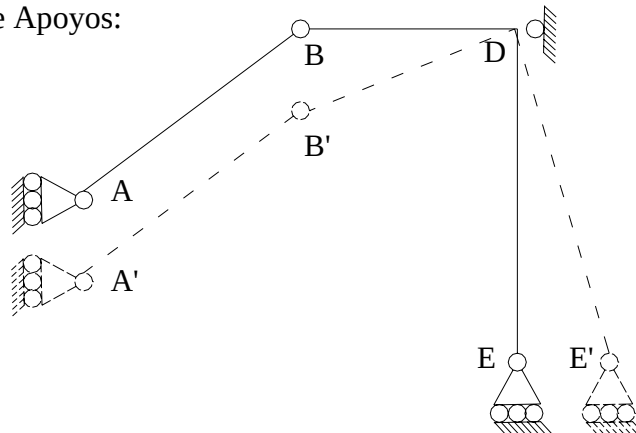
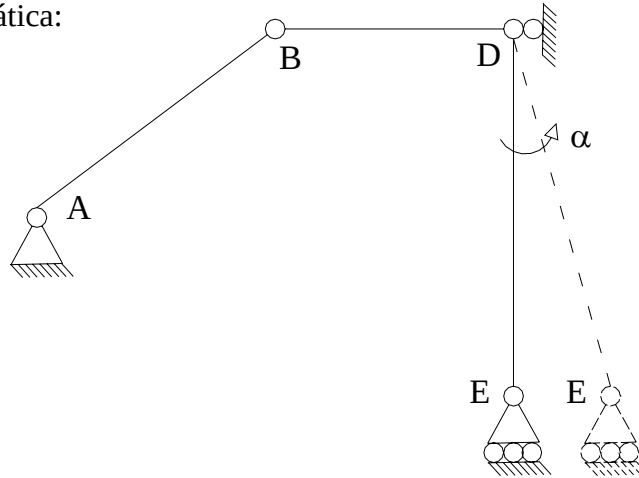
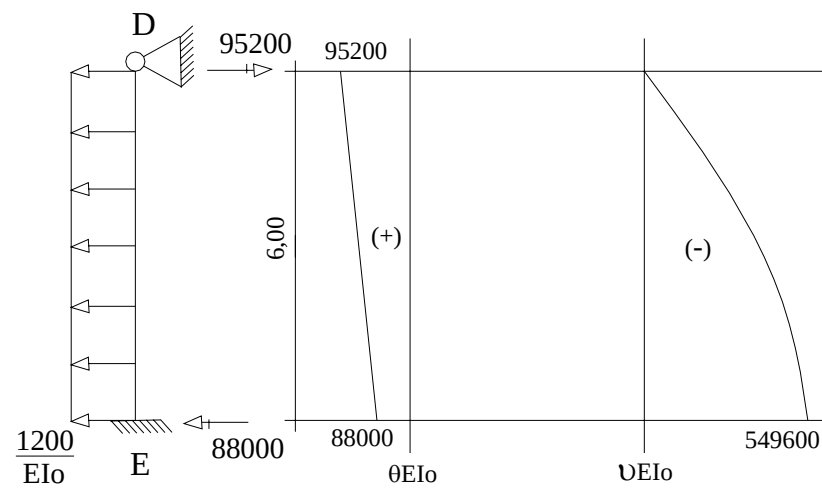
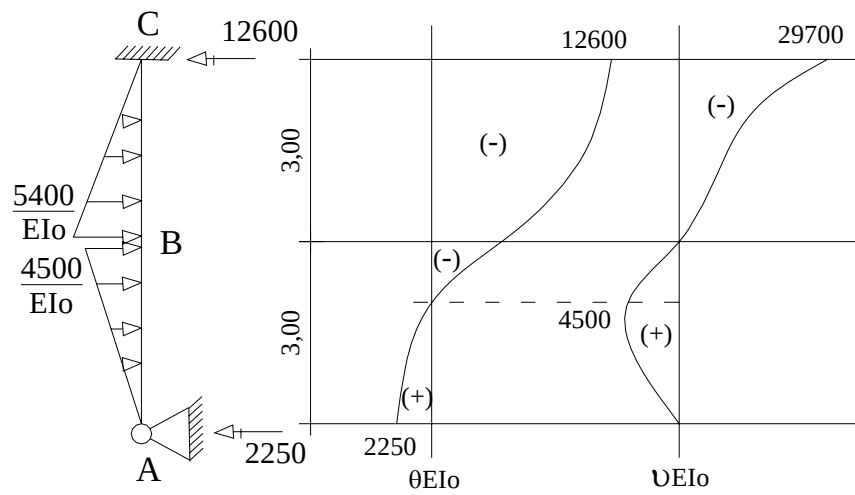
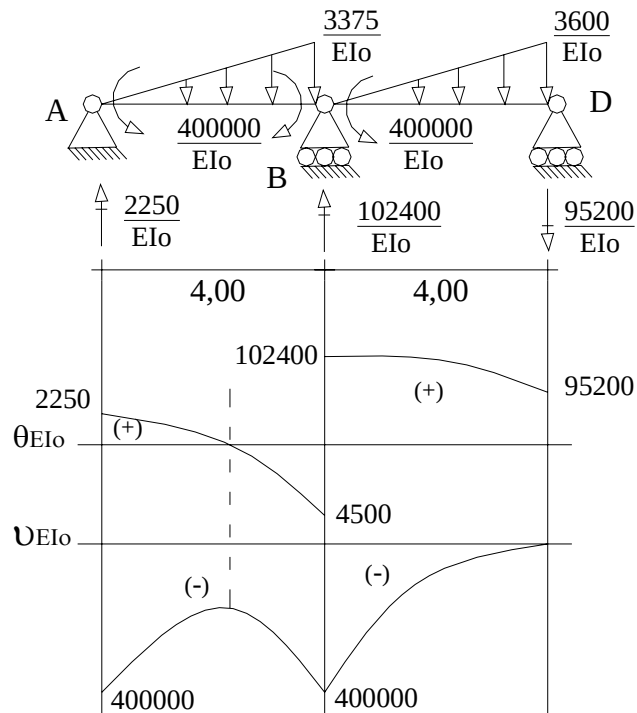


Imagen Cinemática:



Vigas Conjugadas:





10.- Calcular: $S = 3v_D - 5\psi_{AB} - \frac{9}{4}\mu_C$

$$EI_o = 10^6 \text{ Kg.m}$$

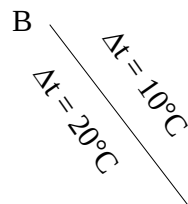
$$\theta_A = 2 \times 10^{-3} \text{ rad}$$

$$\mu_D = -3 \times 10^{-2} \text{ m}$$

$$\alpha t = 1 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$$

$$h_{BC} = 1/2 \text{ m}$$

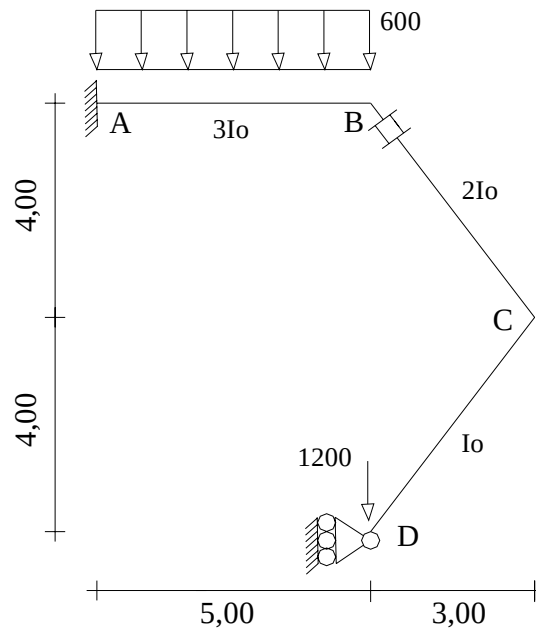
$$A \rightarrow \infty$$



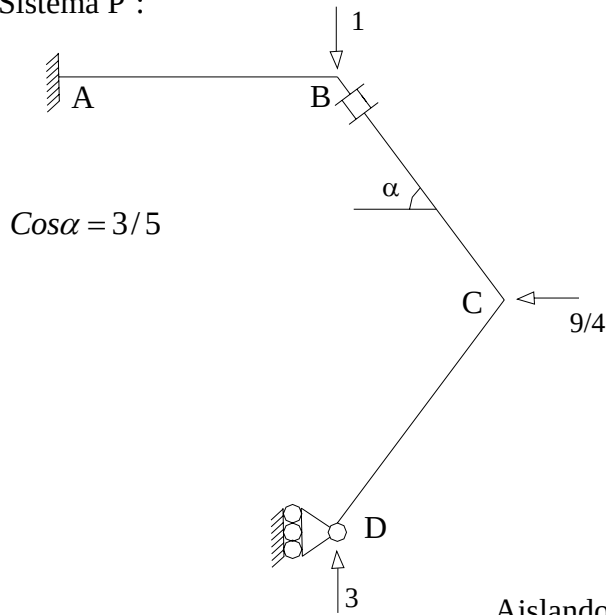
Solución:

$$\Delta t_o = 15^\circ\text{C}$$

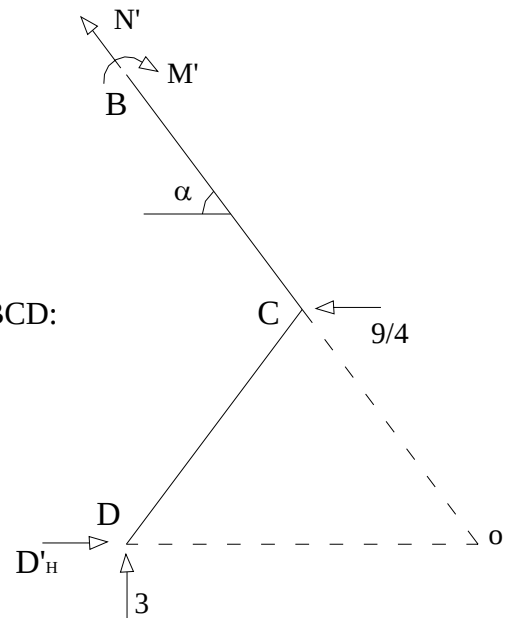
$$\Delta(\Delta t) = 10^\circ\text{C}$$



Sistema P':



Aislando BCD:



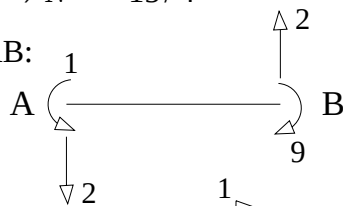
$$\sum M_o = 0 \Rightarrow \frac{9}{4} \times 4 = M' + 3 \times 6$$

$$\Rightarrow M' = -9$$

$$\sum M_D = 0 \Rightarrow N' \cos \alpha \times 8 + \frac{9}{4} \times 4 + 9 = 0$$

$$\Rightarrow N' = -15/4$$

Aislando AB:



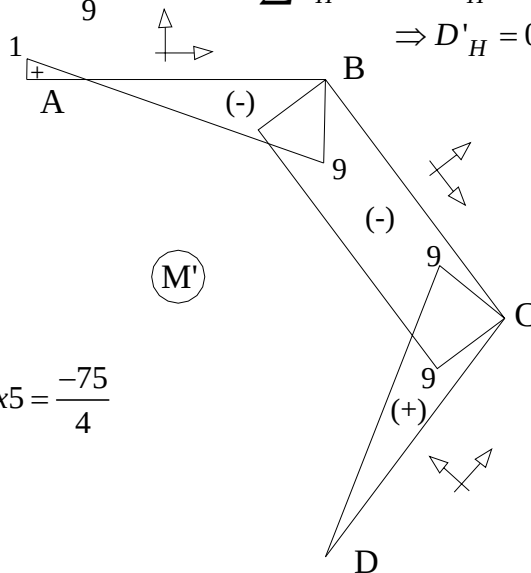
$$\sum F_y = 0 \Rightarrow A'_V = 2 \downarrow$$

$$\sum F_H = 0 \Rightarrow D'_H + N' \cos \alpha = 9/4$$

$$\Rightarrow D'_H = 0$$

$$\Omega_{M'}^{BC} = -45;$$

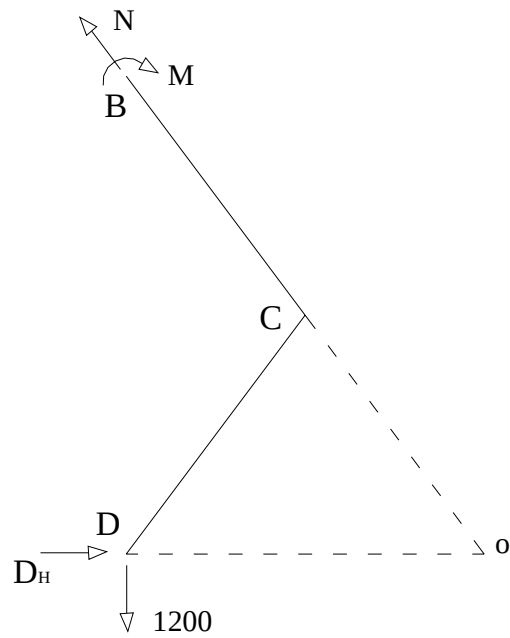
$$\Omega_{N'}^{BC} = \frac{-15}{4} \times 5 = \frac{-75}{4}$$



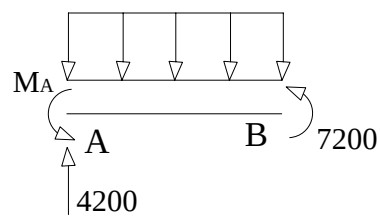
Sistema P:

$$\sum M_o = 0 \Rightarrow M = 7200$$

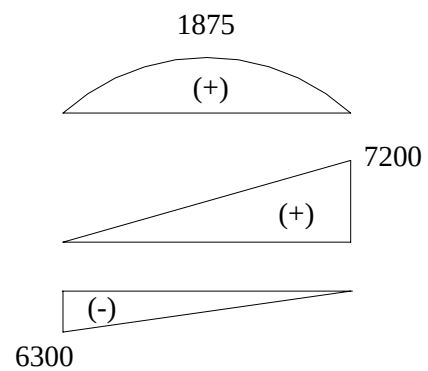
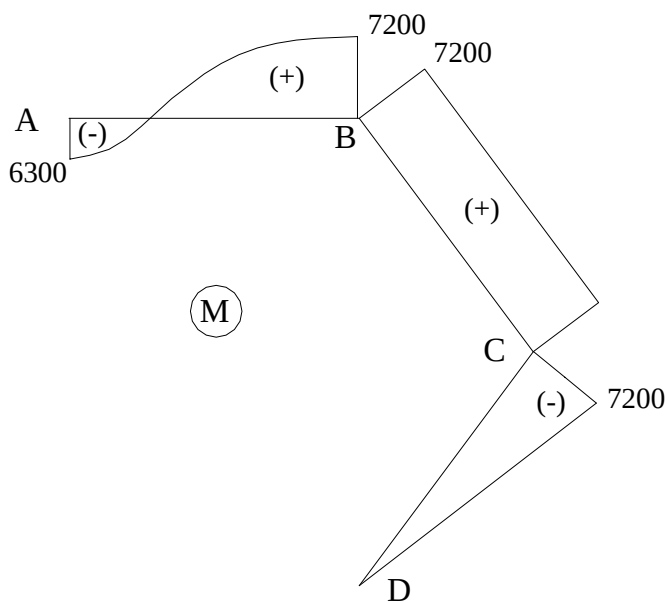
$$\begin{aligned} \sum M_B = 0 &\Rightarrow D_H \times 8 = 7200 \\ &\Rightarrow D_H = 900 \end{aligned}$$



Aislado AB



$$\begin{aligned} \sum M_B = 0 &\Rightarrow M_A + 7200 + 600 \times \frac{5^2}{2} = 4200 \times 5 \\ &\Rightarrow M_A = 6300 \end{aligned}$$



Trabajo Externo:

$$T_{ext} = S - 1 \times 2 \times 10^{-3} \times \frac{10^6}{EI_o};$$

$$T_{ext} = S - \frac{2000}{EI_o}$$

Trabajo Interno:

$$T_i^{AB} = \frac{1}{3} \times \frac{1 \times 1875 \times 5}{3EI_o} + \frac{1}{6} \times \frac{1 \times 7200 \times 5}{3EI_o} - \frac{1}{3} \times \frac{1 \times 6300 \times 5}{3EI_o} - \frac{1}{3} \times \frac{9 \times 1875 \times 5}{3EI_o} -$$

$$- \frac{1}{3} \times \frac{9 \times 7200 \times 5}{3EI_o} + \frac{1}{6} \times \frac{9 \times 6300 \times 5}{3EI_o}$$

$$T_i^{AB} = \frac{-30083,33}{EI_o}$$

$$T_i^{BC} = \frac{-9 \times 7200 \times 5}{2EI_o} + 10^{-5} \times 15 \left(\frac{-75}{4} \right) \frac{10^6}{EI_o} + 10^{-5} \times \frac{10}{1/2} (-45) \frac{10^6}{EI_o}$$

$$T_i^{BC} = -\frac{162000}{EI_o} - \frac{2815,5}{EI_o} - \frac{9000}{EI_o}$$

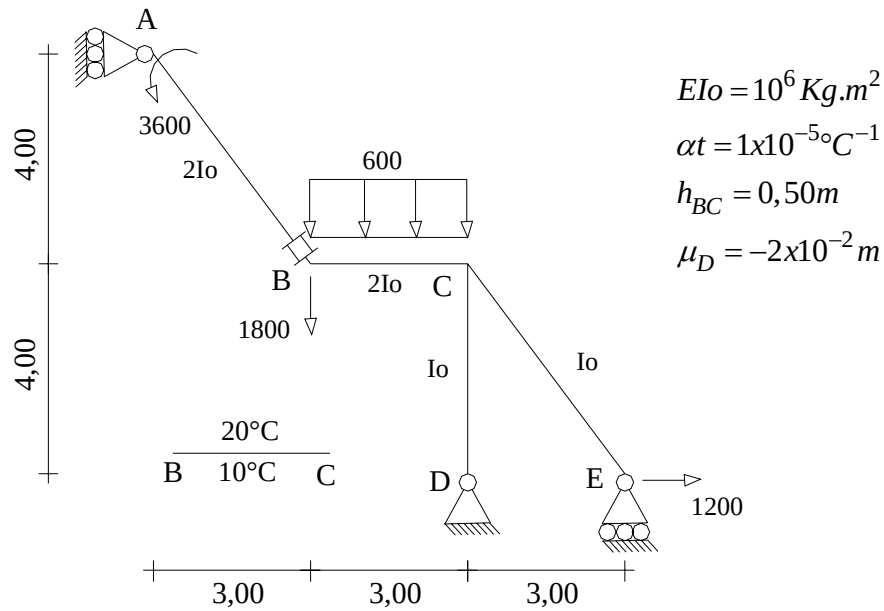
$$T_i^{BC} = -\frac{173812,5}{EI_o}$$

$$T_i^{DC} = -\frac{1}{3} \times \frac{7200 \times 9 \times 5}{EI_o} = -\frac{108000}{EI_o}$$

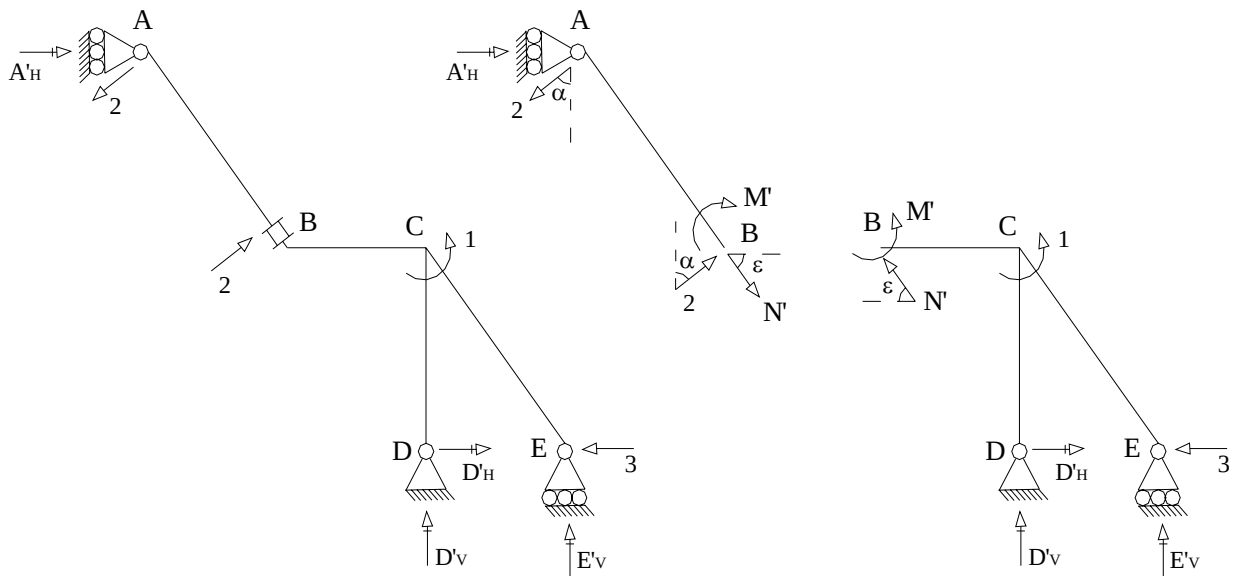
Luego: $T_i^{Total} = -311895,83 / EI_o$;

De donde: $S = -309895,83 / EI_o$

11.- Calcular $S = 10\psi_{AB} - 3\mu_E + \theta_C$



Sistema P':



$$\sum F_V^{AB} = 0 \Rightarrow 2\cos\alpha - 2\cos\alpha - N'\sin\alpha = 0$$

$$\Rightarrow N' = 0$$

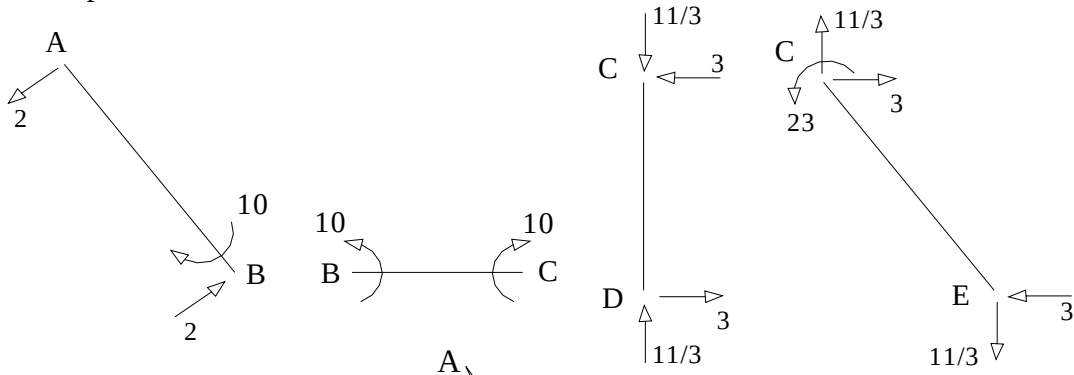
$$\sum F_H^{AB} = 0 \Rightarrow -2\sin\alpha + 2\sin\alpha + A'_V = 0$$

$$\Rightarrow A'_V = 0$$

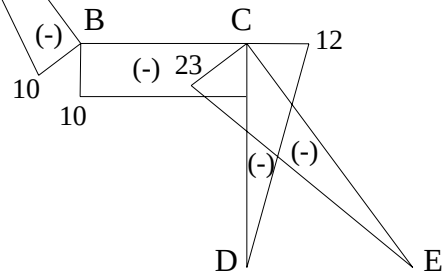
$$\sum M_D^{BCDE} = 0 \Rightarrow 10 + 1 + E'_V \times 3 = 0$$

$$\Rightarrow E'_V = -11/3 \Rightarrow D'_V = 11/3$$

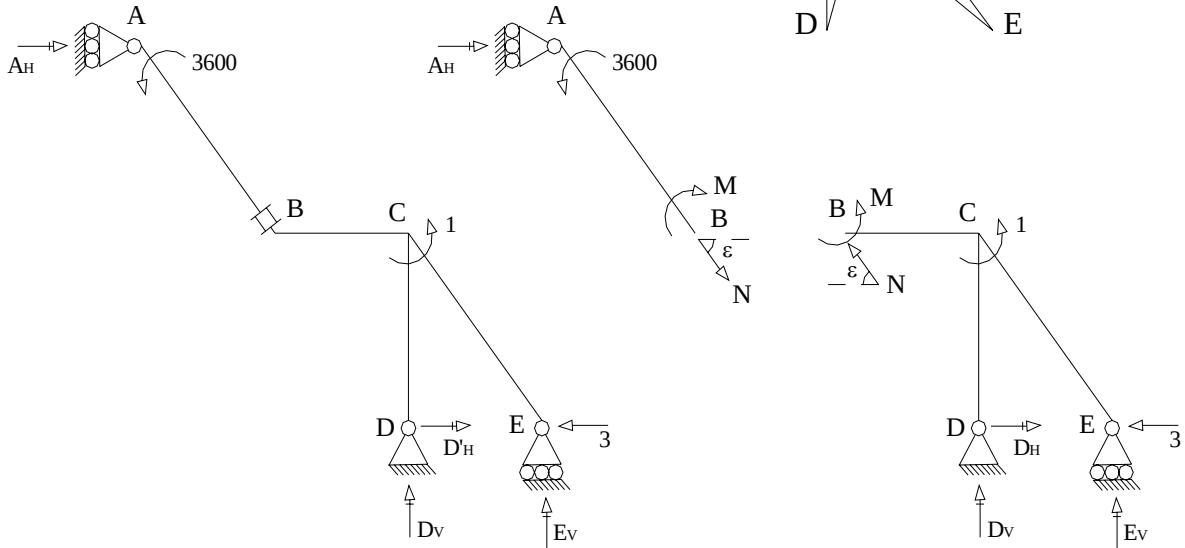
Despiece:



Diagramas de Momentos:



Sistema P:

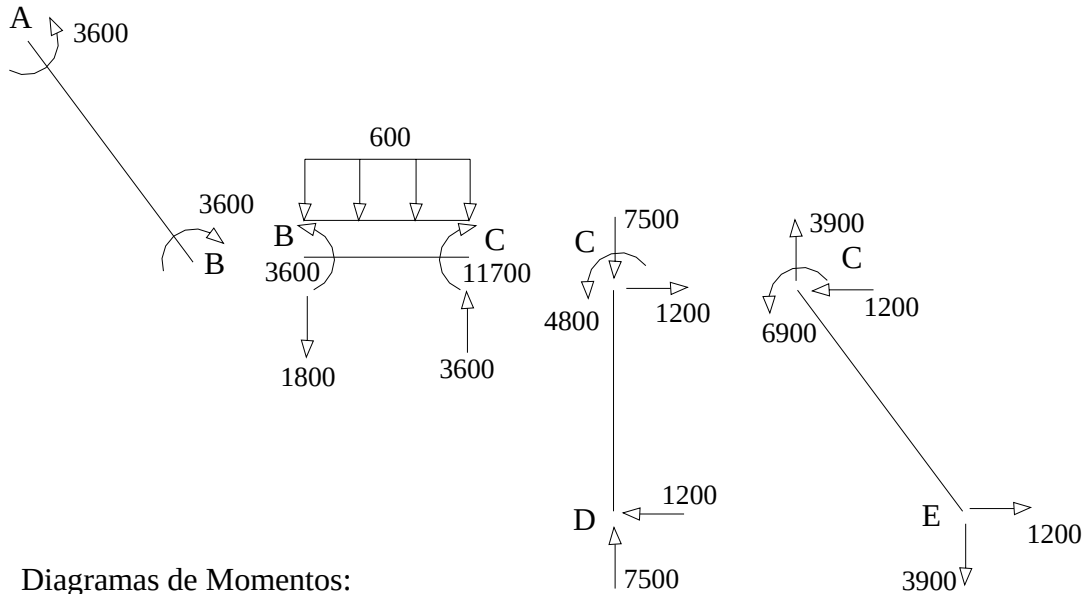


$$\sum M_B^{AB} = 0 \Rightarrow 3600 - 3600 - A_H \times 4 = 0 \Rightarrow A_H = 0 \Rightarrow N = 0$$

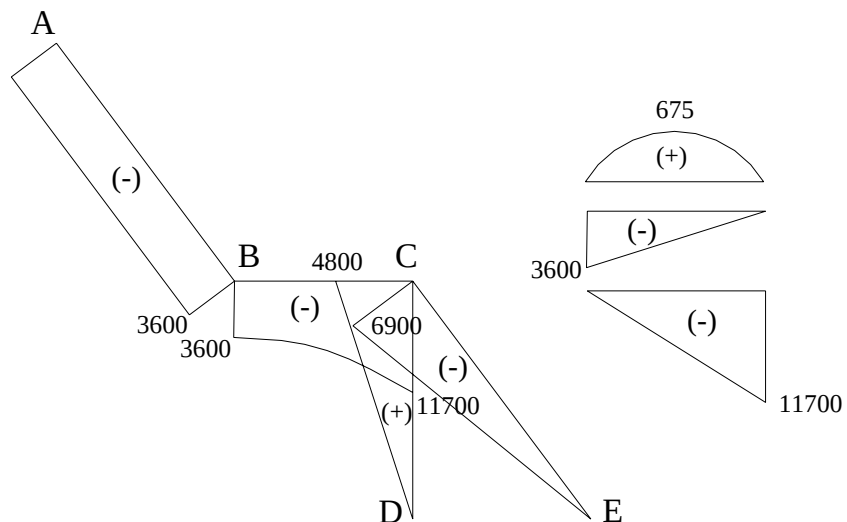
$$\begin{aligned} \sum M_D^{BCDE} &= 0 \Rightarrow 1800 \times 3 + 600 \times 3(3/2) + 3600 + E_V \times 3 = 0 \\ &\Rightarrow 1800 + 900 + 1200 + E_V = 0 \\ &\Rightarrow E_V = -3900 \end{aligned}$$

$$\begin{aligned}\sum F_V^{BCDE} &= 0 \Rightarrow E_V + D_V - 1800 - 600 \times 3 = 0 \\ &\Rightarrow D_V = -E_V + 3600 \\ &\Rightarrow D_V = 7500 \quad (D_H = -1200)\end{aligned}$$

Despiece:



Diagramas de Momentos:



$$T_{Ext} = S + (-3 \times 2 \times 10^{-2} \times \frac{10^6}{EI_o}) = S - \frac{60000}{EI_o}$$

$$T_{Int}^{AD} = \frac{1}{2} \times \frac{10 \times 3600 \times 5}{2EI_o} = \frac{45000}{EI_o}$$

$$T_{Int}^{BC} = \frac{1}{2} \times \frac{3600 \times 10 \times 3}{2EI_o} + \frac{1}{2} \times \frac{11700 \times 10 \times 3}{2EI_o} - \frac{2}{3} \times \frac{675 \times 10 \times 3}{2EI_o} +$$

$$+ 1 \times 10^{-5} \times \frac{10 - 20}{0,5} (-10) 3 \times \frac{10^6}{EI_o} = \frac{114000}{EI_o}$$

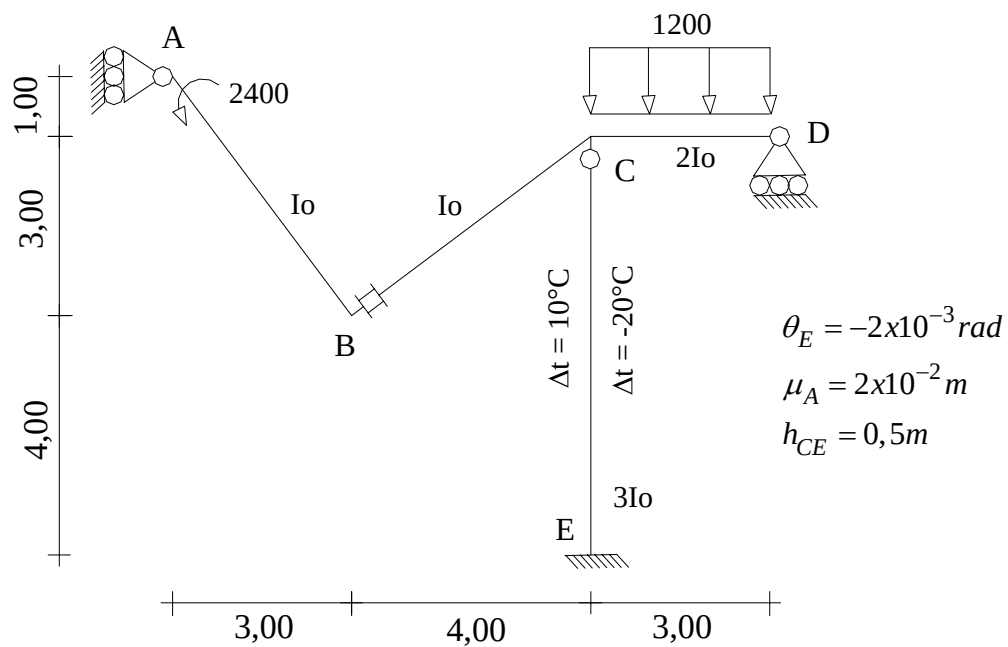
$$T_{Int}^{CD} = \frac{-1}{3} \left(\frac{4800 \times 12 \times 4}{EI_o} \right) = \frac{-76800}{EI_o}$$

$$T_{Int}^{CE} = \frac{1}{3} \left(\frac{6900 \times 23 \times 5}{EI_o} \right) = \frac{264500}{EI_o}$$

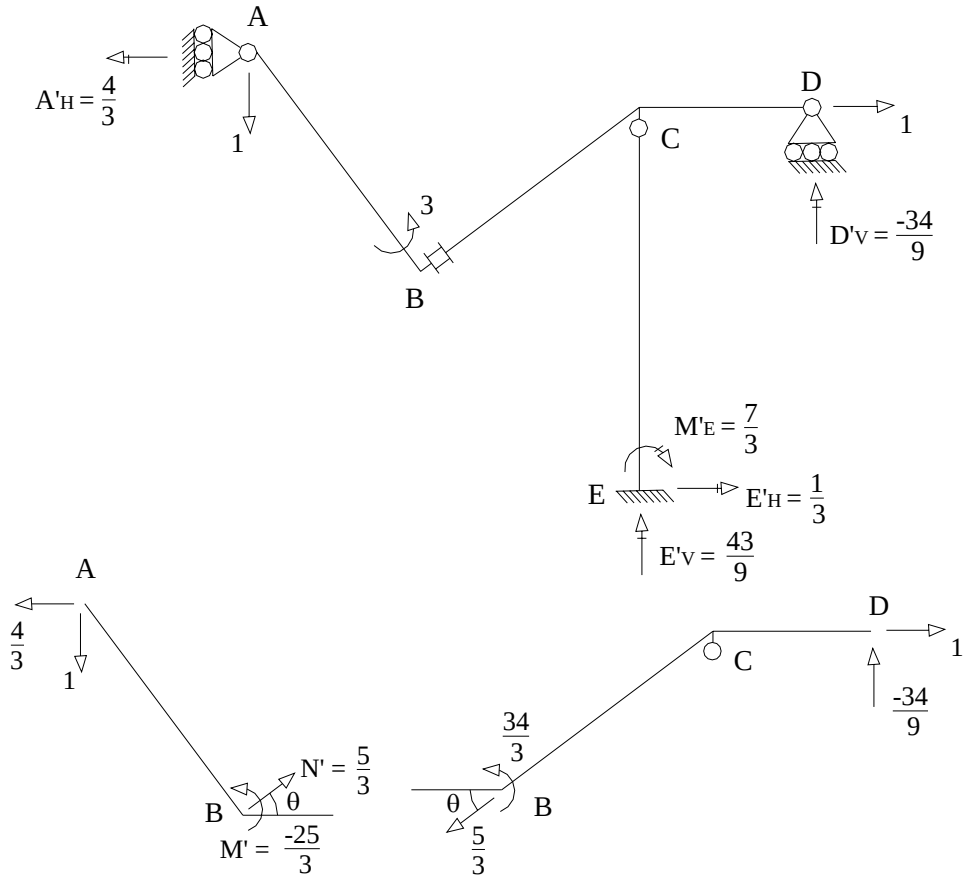
$$T_{Int}^{Total} = \frac{346700}{EI_o}$$

$$S = \frac{406700}{EI_o}$$

12.- Calcular: $S = 3\theta_B - \nu_A + \mu_D$



Sistema P':



$$\begin{aligned}\sum M_B^{AB} &= 0 \Rightarrow N' \text{Sen}\theta = 1 \\ \Rightarrow N' &= \frac{1}{\text{Sen}\theta} = \frac{5}{3}\end{aligned}$$

$$\begin{aligned}\sum F_H^{AB} &= 0 \Rightarrow A'_H = N' \text{Cos}\theta \\ \Rightarrow A'_H &= (5/3) \times (4/5) = 4/3\end{aligned}$$

$$\begin{aligned}\sum M_A^{AB} &= 0 \Rightarrow M' + 5/3 \text{Cos}\theta(4) + 5/3 \text{Sen}\theta(3) = 0 \\ \Rightarrow M' + (5/3) \times (4/5) \times 4 + (5/3) \times (3/5) \times 3 &= 0 \\ \Rightarrow M' + 16/3 + 9/3 &= 0 \\ \Rightarrow M' &= -25/3\end{aligned}$$

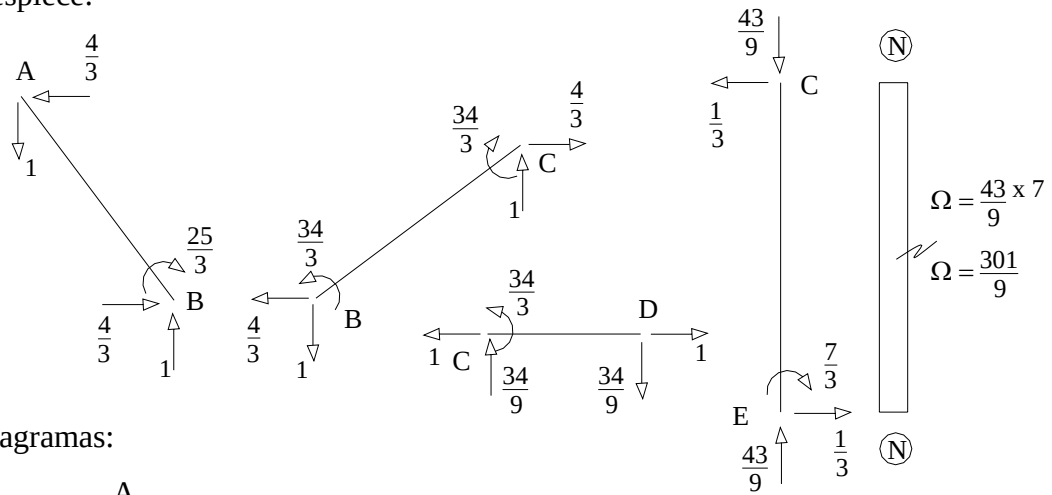
$$\begin{aligned}\sum F_V &= 0 \Rightarrow -1 + E'_V - 34/9 = 0 \\ \Rightarrow E'_V &= 34/9 + 1 \\ \Rightarrow E'_V &= 43/9\end{aligned}$$

$$\begin{aligned}\sum F_H &= 0 \Rightarrow -4/3 + E'_H + 1 = 0 \\ \Rightarrow E'_H &= 4/3 - 1 \\ \Rightarrow E'_H &= 1/3\end{aligned}$$

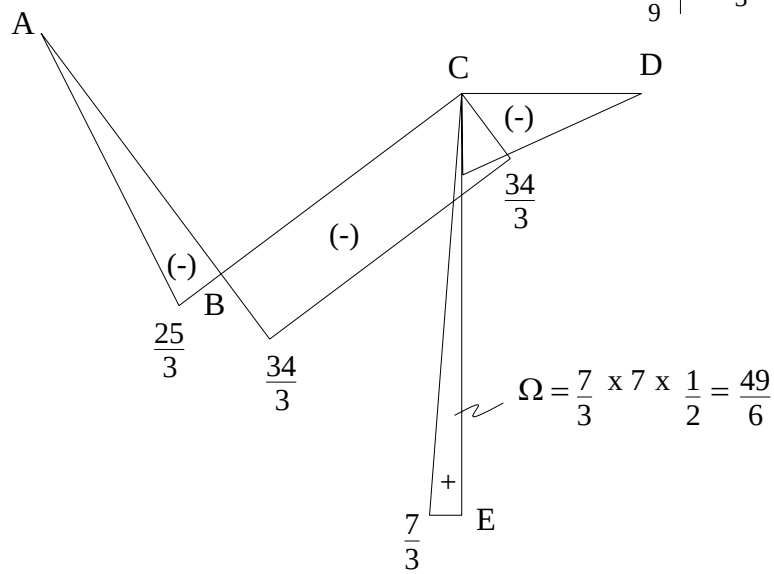
$$\begin{aligned}\sum M_C^{BCD} &= 0 \Rightarrow 34/3 + D'_V \times 3 = 0 \\ \Rightarrow D'_V &= -34/9\end{aligned}$$

$$\begin{aligned}\sum M_C^{CE} &= 0 \Rightarrow 1/3 \times 7 - M'_E = 0 \\ \Rightarrow M'_E &= 7/3\end{aligned}$$

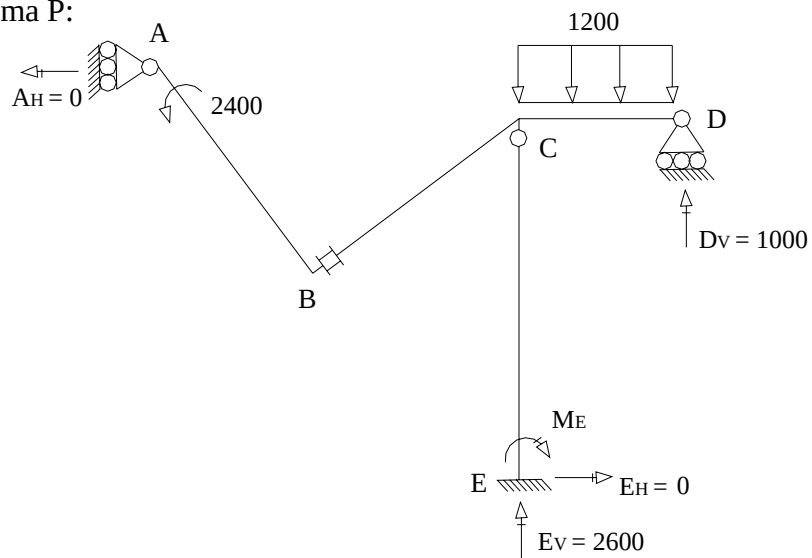
Despiece:

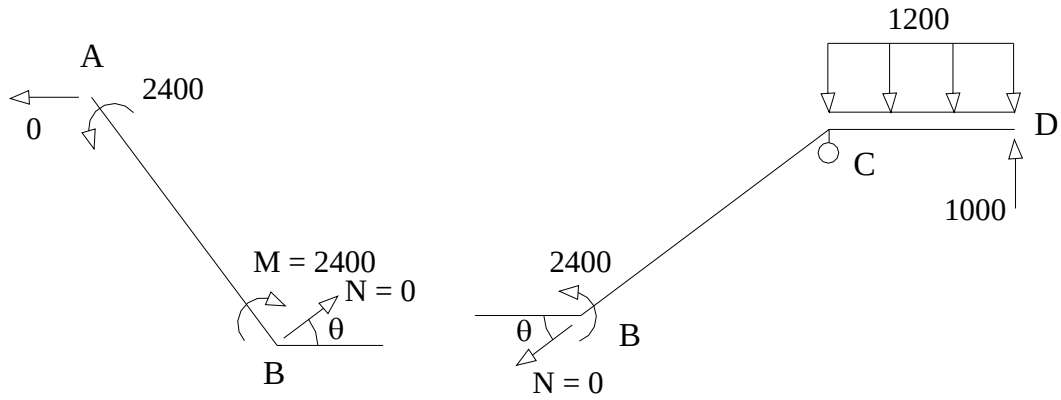


Diagramas:



Sistema P:





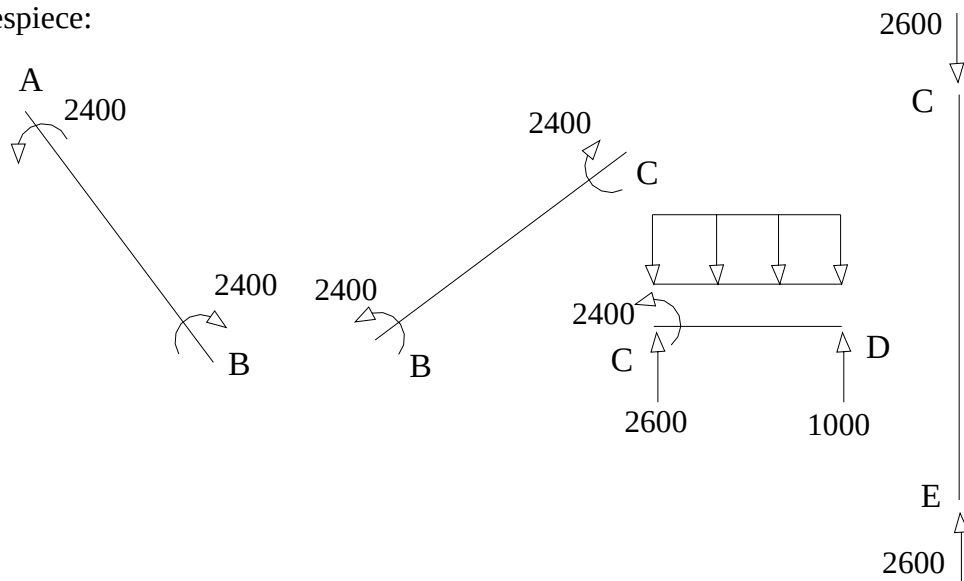
$$N = 0 \Rightarrow A_H = 0 \Rightarrow E_H = 0 \Rightarrow M_E = 0$$

$$\sum M_B^{AB} = 0 \Rightarrow M = 2400$$

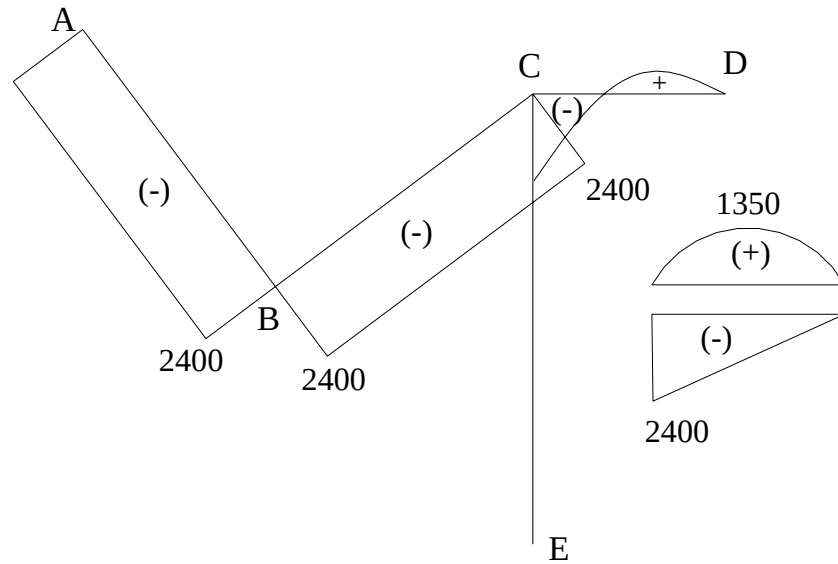
$$\begin{aligned} \sum M_C^{BCD} = 0 &\Rightarrow 2400 - 1200 \times 3 \times 1,5 + D_V(3) = 0 \\ &\Rightarrow 2400 - 3600 \times 3/2 + 3D_V = 0 \\ &\Rightarrow D_V = 1000 \end{aligned}$$

$$\begin{aligned} \sum F_V^{Toda} = 0 &\Rightarrow E_V + 1000 - 1200 \times 3 = 0 \\ &\Rightarrow E_V = 2600 \end{aligned}$$

Despiece:



Diagramas:



$$T_{Ext} = S + 7/3 \times 2 \times 10^{-3} - 4/3 \times 2 \times 10^{-2} = S - 22000 / EI_o$$

$$T_{Int}^{AB} = \frac{1}{2} \left[\frac{(25/3) \times 2400 \times 5}{EI_o} \right] = \frac{50000}{EI_o}$$

$$T_{Int}^{BC} = \frac{(34/3) \times 2400 \times 5}{EI_o} = \frac{136000}{EI_o}$$

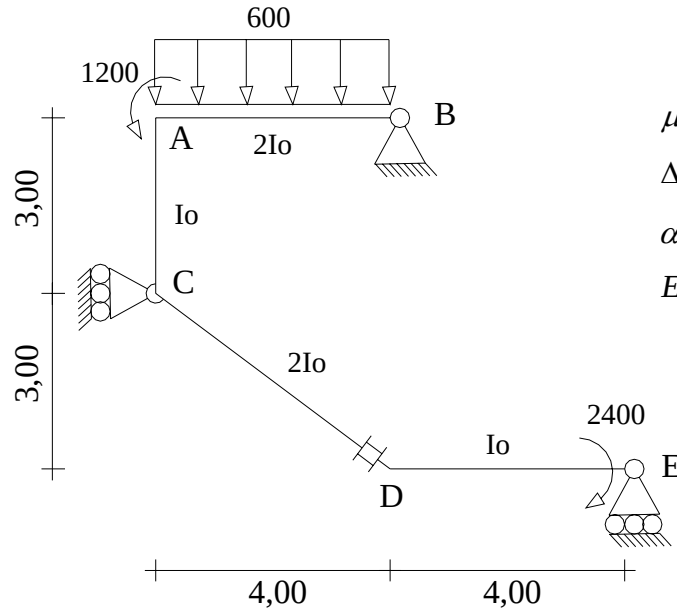
$$T_{Int}^{CD} = \frac{1}{3} \left[\frac{(34/3) \times 2400 \times 3}{2EI_o} \right] - \frac{1}{3} \left[\frac{(34/3) \times 1350 \times 3}{2EI_o} \right] = \frac{5950}{EI_o}$$

$$T_{Int}^{EC} = \frac{(1 \times 10^{-5})(-20 - 10)(49/6)}{0,5} \times \frac{10^6}{EI_o} + (1 \times 10^{-5}) \times \frac{10 - 20}{2} \times \frac{-301}{9} \times \frac{10^6}{EI_o} = \frac{-3227,7}{EI_o}$$

$$T_{Int}^{Total} = \frac{188722,22}{EI_o}$$

$$S = \frac{210722,22}{EI_o}$$

13.- Calcular: $S = 2\mu_E - 10\psi_{CD} - 3v_C$



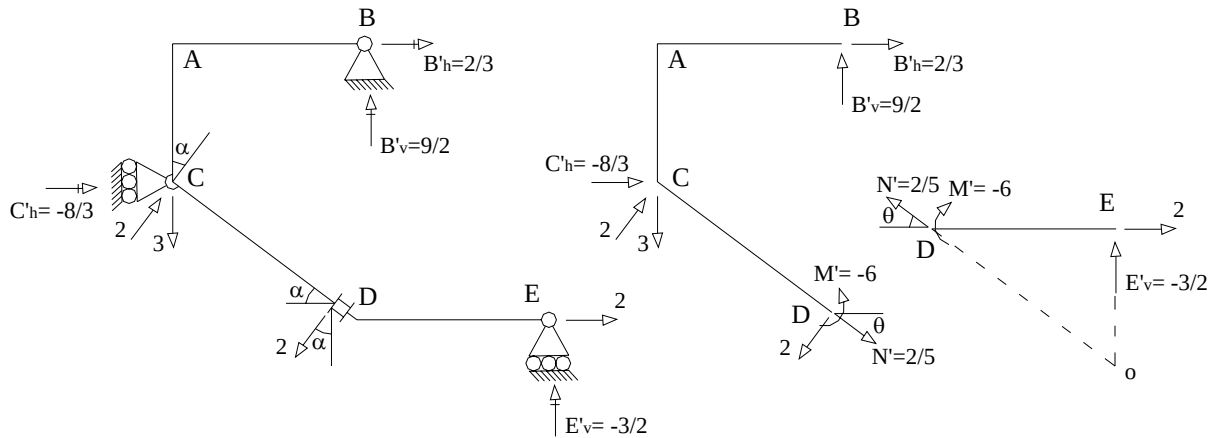
$$\mu_C = -2 \times 10^{-2} m$$

$$\Delta t_{CD} = 10^\circ C$$

$$\alpha t = 1 \times 10^{-5} {}^\circ C^{-1}$$

$$EI_0 = 10^6 Kg.m^2$$

Sistema P':



$$\sum M_o^{DE} = 0 \Rightarrow -M' - 2 \times 3 = 0$$

$$\Rightarrow M' = -6$$

$$\sum F_V^{DE} = 0 \Rightarrow -N' \text{Sen} \theta = E'_v$$

$$\Rightarrow E'_v = \frac{-5}{2} \times \frac{3}{5} = \frac{-3}{2}$$

$$\sum F_H^{DE} = 0 \Rightarrow N' \text{Cos} \theta = 2$$

$$\Rightarrow N' = \frac{2}{\text{Cos} \theta} = \frac{10}{4} = \frac{5}{2}$$

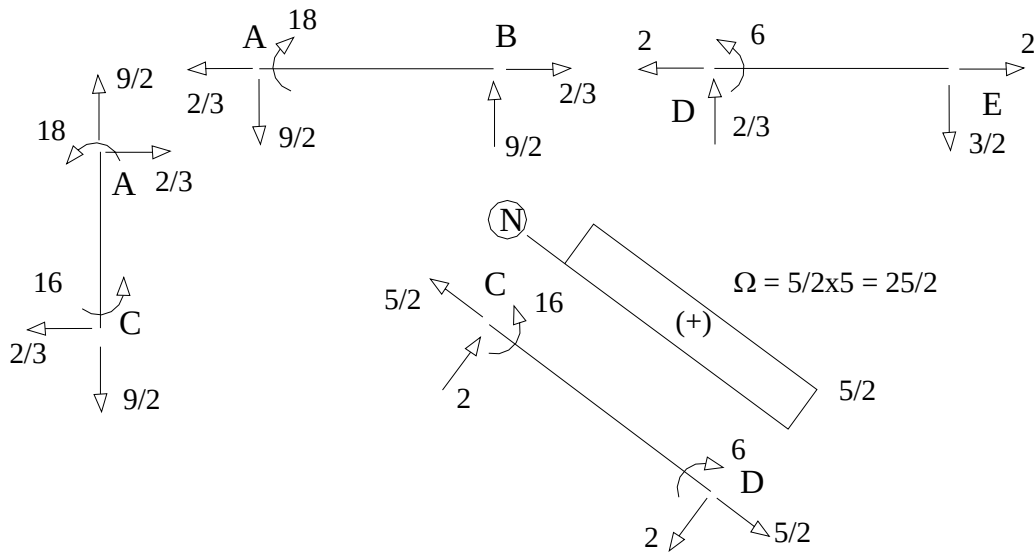
$$\sum F_V^{Toda} = 0 \Rightarrow B'_v - 3/2 - 3 = 0$$

$$\Rightarrow B'_v = 3/2 + 3 = 9/2$$

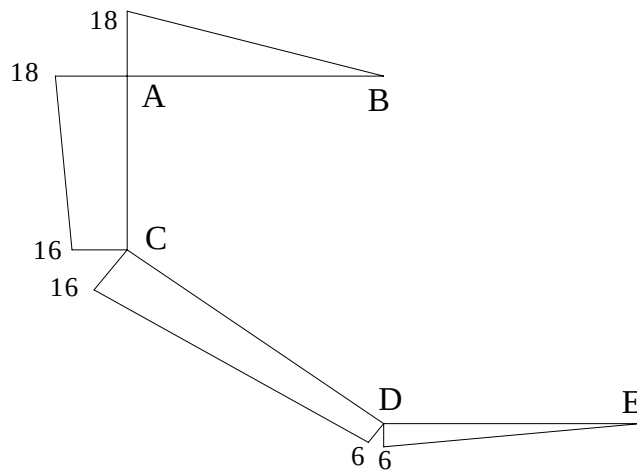
$$\begin{aligned}
 \sum M_C^{Toda} = 0 &\Rightarrow -2\cos\alpha(4) - 2\sin\alpha(3) - B'_H \times 3 + (9/2) \times 4 + 2 \times 3 + E'_V \times 8 = 0 \\
 &\Rightarrow -8(4/5) - 6(3/5) - B'_H \times 3 + 18 + 6 - (3/2)8 = 0 \\
 &\Rightarrow \frac{-32}{5} - \frac{18}{5} - B'_H \times 3 + 24 - 12 = 0 \\
 &\Rightarrow -B'_H + 2 = 0 \\
 &\Rightarrow B'_H = 2/3
 \end{aligned}$$

$$\begin{aligned}
 \sum F_h^{Toda} = 0 &\Rightarrow 2/3 + C'_H + 2 = 0 \\
 &\Rightarrow C'_H = -8/3
 \end{aligned}$$

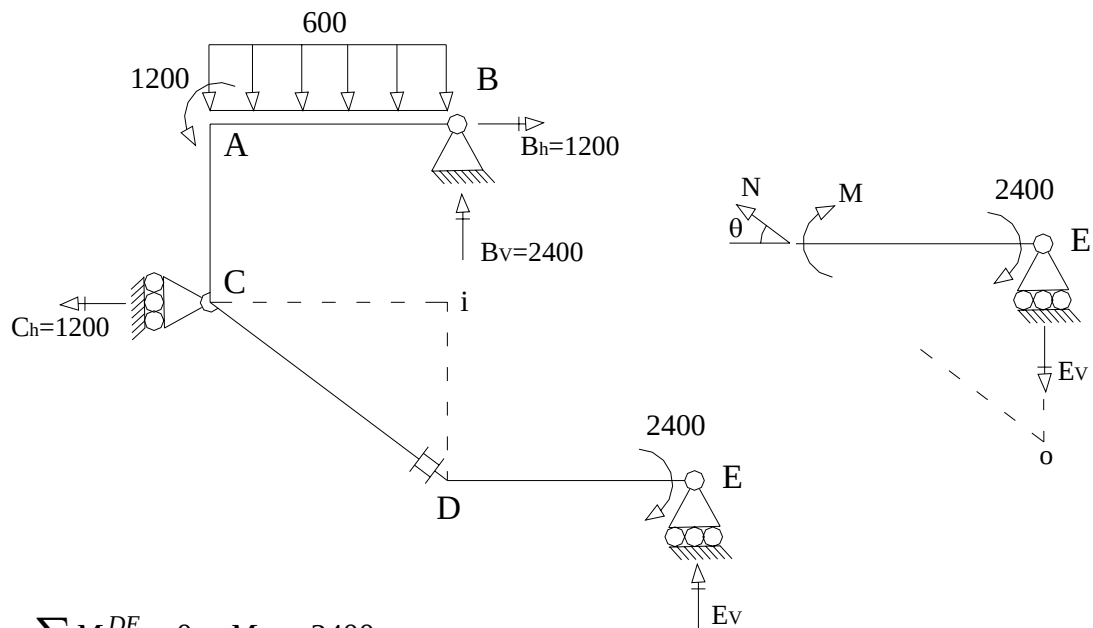
Despiece:



Diagramas:



Sistema P:



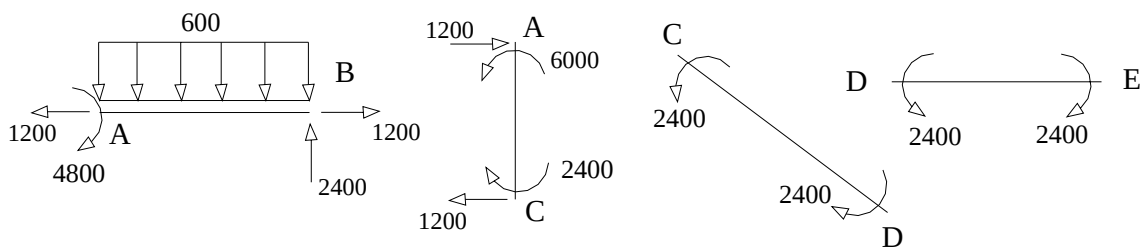
$$\sum M_o^{DE} = 0 \Rightarrow M = -2400$$

$$\sum F_H^{DE} = 0 \Rightarrow N = 0; E_V = 0$$

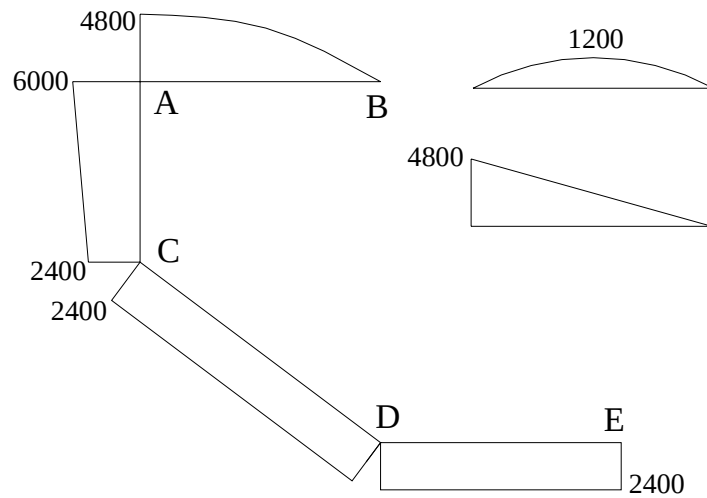
$$\begin{aligned} \sum M_i^{Toda} = 0 &\Rightarrow 1200 + 600(4)(2) - B_H(3) - 2400 = 0 \\ &\Rightarrow B_H(3) = 3600 \\ &\Rightarrow B_H = 1200 \end{aligned}$$

$$\begin{aligned} \sum F_V^{Toda} = 0 &\Rightarrow B_V - 600(4) = 0 \\ &\Rightarrow B_V = 2400 \end{aligned}$$

Despiece:



Diagramas:



$$T_{Ext} = S + \left(\frac{8}{3} \times 2 \times 10^{-2} \times \frac{10^6}{EI_o} \right)$$

$$T_{Int}^{AB} = \frac{1}{3} \left(\frac{1200 \times 18 \times 4}{2EI_o} \right) + \frac{1}{3} \left(\frac{18 \times 4800 \times 4}{2EI_o} \right) = \frac{72000}{EI_o}$$

$$T_{Int}^{AC} = \frac{3}{6} \left(\frac{2 \times 16 \times 2400 + 2 \times 18 \times 6000 + 16 \times 6000 + 18 \times 2400}{EI_o} \right) = \frac{216000}{EI_o}$$

$$T_{Int}^{CD} = \frac{(6+16)2400 \times 5}{4EI_o} + (1 \times 10^{-5})10 \left(\frac{25}{2} \right) \frac{10^6}{EI_o} = \frac{67250}{EI_o}$$

$$T_{Int}^{DE} = \frac{2400 \times 6 \times 4}{2EI_o} = \frac{28800}{EI_o}$$

$$T_{Int}^{Total} = \frac{384050}{EI_o}$$

$$S = \frac{384050}{EI_o} - \frac{53333,33}{EI_o} = \frac{330716,67}{EI_o}$$